The Spaghetti Engine
VISUAL SYSTEM MODELLING WORKSPACE

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User Manual: Building, Simulating, and Analyzing Fuzzy Cognitive Maps

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1. Introduction

The Spaghetti Engine is a single-page, in-browser tool for building, simulating, and analyzing Fuzzy Cognitive Maps (FCMs): causal models in which system variables ("concepts") influence one another through signed, weighted relationships. It was built to support participatory system modelling for research, letting a researcher or a group of stakeholders sketch out what drives a system, then interrogate that structure mathematically instead of just visually.

Since version 0.2 the tool is built for nexus analyses in general: systems made up of several interacting subsystems, such as water, energy, food, ecosystems, or policy domains. Concepts can be grouped into modules (subsystems), and every analysis can be run on the whole system, on one module, on a combination of modules, or on the interface between them. Since version 0.2.1 each module can also be folded into a single block on the map, so the model can be read as a set of sub-models: an overview of how the modules are linked, one module opened up inside the rest, or the complete model. Version 0.3.0 adds Transition Pathway Analysis: it looks for the order in which interventions are best implemented, and compares the resulting pathways on what they achieve, how much effort they need, and when the benefits arrive. Version 0.4.0 adds a Monte Carlo tab, which tests how far the results depend on the uncertainty in the weights; lets every simulation run without a squashing function, so that the raw dynamics of the map can be explored; groups the tabs by what you are doing; reworks the Network tab for larger models and for moving relationships by hand; and adds figure export at publication quality. Version 0.4.5 lets the network map itself serve as a figure of the system, and adds smart Auto Arrange layouts that keep crossings down, group tightly linked concepts, and can place drivers on the left and outcomes on the right.

The tool covers the full workflow a modelling exercise like this normally requires:

- Building a model visually (a drag-and-drop network canvas with a guided Inspector) or in bulk (a spreadsheet-style table, or pasting an adjacency matrix straight out of Excel), with an automatic Model check that points out likely mistakes, and a definition for every concept.
- Grouping concepts into modules, folding modules into sub-models on the map, and choosing the analysis scope: the full nexus, one or more modules, or only the relationships between modules (the nexus interface).
- Analyzing the model's structure (which concepts are hubs, drivers, receivers, or brokers) with no simulation required, and comparing every concept's importance across scales.
- Tracing influence routes: the chains of cause and effect that connect concepts inside and between modules, the concepts those chains depend on, and the feedback loops in the system.
- Running the model to see where it settles on its own (its baseline equilibrium), from the model's own starting values or from a custom starting state. The transfer function can also be switched off, to see the model's raw linear dynamics.
- Building "what if" scenarios and comparing the result against the baseline.
- Sensitivity analysis: sweeping one concept across its full range and watching how every other concept responds.
- **Monte Carlo simulation:** re-running the model many times with the relationship weights (and, if you wish, the starting values) nudged at random, to see how far a result depends on the uncertainty behind the weights.
- Transition point analysis: finding the intervention intensity at which a concept's response changes fastest.
- Transition pathway analysis: combining interventions into ordered sequences, each implemented up to its transition point, and comparing the pathways on gain, effort, and timing.
- **Publication-quality figures:** a figure panel on the main charts sets the size in millimeters, the resolution, fonts, palette, and legend, and exports the chart as PNG, SVG, or PDF. The network map has a Figure style of its own, exported at three times screen resolution.

- Exporting everything into one documented Excel workbook that states which part of the system each result describes, and exporting the model itself as tables (concepts with definitions, relationships, adjacency matrix).

> **Prototype scope:** The tool runs entirely in your browser. Your model is saved automatically in that browser and restored on your next visit; use Export JSON to keep a file copy or to move a model to another computer. Analysis results are not saved and are re-run on demand. Exporting a whole analysis directly to PDF/DOCX/PPTX from within the tool is on the roadmap but not implemented yet; individual charts can be exported as PNG, SVG, or PDF (Section 14.2).

1.1 Who this manual is for

This manual assumes no prior experience with the tool and only a little familiarity with Fuzzy Cognitive Mapping in general. Section 2 gives a short primer. It is written for:

- Researchers building and refining an FCM model of a system or nexus.
- Workshop facilitators and participants who co-construct a model with stakeholders and want to understand what the tool is showing them.
- Readers of work this tool supports, who want to understand exactly how a reported number (an influence score, an equilibrium, an elasticity, a route strength) was computed.

1.2 What is new in version 0.4.5

Version 0.4.5 reworks the Network tab, so that a map can serve as a figure of the system:

Area	What changed	Where
Figure style	A new Figure / Editing switch. Figure style (the default) shows name-only cards, thin lines that are dashed when decreasing, lines that bow around cards in their way, and weight labels placed clear of cards and of each other. Editing style is the earlier look, with each concept's activation on its card.	Section 4.1
Smart Auto Arrange	Four new layouts: Readable (the default), Systems thinking, Cluster view, and Centrality view. They group tightly linked concepts and then remove crossings, overlapping cards, and clashing labels. Cards glide to their new places, a line under the toolbar reports the result, and one undo restores the previous arrangement.	Section 4.1, Auto Arrange
Export PNG	In Figure style, the whole map, cropped to its content, at three times screen resolution, without the zoom controls, minimap, or connection dots, whatever the current zoom.	Sections 4.1 and 14.2

Version 0.4.4 changed the Transition Pathways tab: an outcome weight or flat-response threshold left blank now takes its default value, the analysis accepts expert-defined sequences, and the robustness test also covers outcome weights and archetype thresholds. The development log of 0.4.4 describes these changes.

Version 0.4.3 changed how transition points are defined, on both transition tabs:

Area	What changed	Where
Transition point	The transition point is now the intensity from which the marginal benefit declines most steeply: the point of diminishing returns, from where each further step of effort adds markedly less. Before, it was the steepest step of the response. A transition point of 1 means that the marginal benefit does not decline before full intensity.	Sections 12 and 13
Response types	Early, Middle, Late, and Proportional replace Front-Loaded, Threshold-Type, and Back-Loaded; Flat is unchanged.	Sections 12.2 and 13.6
Change and Share at TP	Change at TP is now the change up to the transition point itself, Y(TP) - Y(0), and a new column, Share at TP, gives it as a share of Max Change.	Section 12.5
Precision of the runs	Every run on both transition tabs continues until no concept changes by more than 1e-9 per iteration, with at least 1000 iterations, and the tabs report how many runs converged and how many reached this precision.	Sections 12.6 and 13.2
Transition Pathways	Each stage stops at the lever's marginal-benefit transition point of the total outcome gain, and a lever is only added if the gain there is higher than the gain already reached. Pathway results differ from those of 0.4.2.	Section 13

Version 0.4.2 made the tool easier to read and added one result column:

Area	What changed	Where
Readability	Text on light backgrounds is black or very dark grey instead of light grey, and the small text sizes are larger (the smallest text is now 11.5 pixels). Chart axis labels are darker and larger.	Section 3.1
Change at TP	The Transition Point results table, its CSV export, and the Analysis Workbook show how far each outcome has moved once the steepest step, which begins at the transition point, is complete, next to Max Change, the change at full intensity.	Sections 12.5 and 14

Version 0.4.1 changed one convention, the direction of a decrease:

Area	What changed	Where
Transition Point direction	A decrease now sweeps the intervention from 0 down to -1 instead of from 1 down to 0, like a scenario in which the concept is a driver at -1. An increase still sweeps it from 0 up to +1. The intensity axis still runs from 0 to 1: for a decrease, an intensity of 0.4 holds the intervention at -0.4. The last step of each sweep is the same run as the matching scenario.	Section 12
Transition Pathways direction	A decreasing lever follows the same rule: intensity 0 to 1 holds it at 0 to -1. Before, intensity 0 held a decreasing lever at +1 and full intensity at 0, so pathway results for decreasing levers differ from those of 0.4.0. Pathways with only increasing levers are unchanged.	Section 13
Labels and exports	The buttons, headings, explanations, and the Direction columns of the exports say Decrease (0 to -1).	Sections 12, 13, and 14

Version 0.4.0 introduced the following:

Area	What changed	Where
Tab groups	The ten tabs are arranged in four groups, so the tabs a participatory session needs (Model editor, Network, Influence Metrics, Influence Routes) sit apart from the analyst's tools.	Section 3.1
Monte Carlo (new tab)	Re-runs the model many times with every weight nudged at random, to show how far each result depends on the uncertainty behind the weights: a summary table, histograms, and a seed that makes a run exactly repeatable.	Section 11
Run without a squashing function	A fourth choice next to hyperbolic tangent, sigmoid, and trivalent: the plain linear update with no bound, to explore the model's raw dynamics. A run that grows without limit is stopped and reported as diverging.	Sections 2.2, 8.1, 9.5
Network tab	A larger canvas with a collapsible Inspector; hovering a concept isolates its connections; a search box; relationships shown by arrow color and width only; drag to bend and to reconnect relationships, with snapping to a visible anchor point; smoother dragging and hovering.	Section 4.1
Publication-quality figures	Customize & export on the main charts: size in millimeters, resolution, fonts, palettes (including a color-blind-safe and a grayscale one), legend, which series appear and in what order; export as PNG, SVG, or PDF.	Section 14.2
Clearer tables	Every column of the Transition Point results table sorts, and Range and Efficiency columns are added; the Sensitivity impact ranking and elasticity tables are tidier and export as CSV.	Sections 10.2, 12.5
Pathway charts	The effort-gain plane labels the selected pathway, the benefit accrual chart shows the Early Benefit Index as a filled area, and the pathway charts follow the figure panel. The gain measure is called Total Outcome Gain (TOG).	Section 13
Peat meadow example	A third example model: the Dutch peat meadow case study, with five levers and seven scenarios.	Section 3.6
About & documentation	A page in the tool that says what it is for, who it is for, and why it was made; offers this documentation as PDF downloads; and gives the contact details.	Section 3.1

Version 0.3.0 introduced the following:

Area	What changed	Where
Transition Pathways (new tab)	Combines intervention levers into ordered pathways. Each stage implements one lever up to its transition point in the state the earlier stages have created, so the tab shows how interventions make each other more or less effective. Pathways are compared on seven metrics, sorted into four archetypes (Low-Hanging Fruit, High-Impact, Efficiency, Deep Investment), and ranked.	Section 13

Area	What changed	Where
Effort-gain plane and portfolios	A chart of every pathway by effort and gain with the efficient frontier, and implementation portfolios that show which opening moves keep the best pathways within reach.	Section 13.6
Checks on the results	An invariance test shows whether the order of the levers changes where the system ends up, and an optional robustness test repeats the analysis under four other simulation settings.	Section 13.6
Exports	CSV and PNG exports on the tab, and eight new sheets in the Analysis Workbook when a pathway analysis has been run.	Sections 13.7, 14.3

Version 0.2.1 introduced the following:

Area	What changed	Where
Foldable modules	Each module can be folded into one block on the Network tab. A structure switch shows the full model, a module overview (every module folded, with bundled links between them), or one module opened up with the others folded around it.	Section 4.1
Concept definitions	Every concept has a definition field, in the Model editor and in the Inspector. Definitions are included in the exports.	Sections 4.1, 4.2
Model tables export	Concepts with definitions, relationships, and the adjacency matrix as CSV files or as one Excel workbook, from the Model editor.	Section 4.2
Replacing a model	Starting a blank model, loading an example, importing, and building from a matrix now ask through a dialog in the tool itself, with an option to download a copy first. This fixes a problem where some browsers silently blocked the question, so the model could not be changed.	Section 3.3
New example model	The Sustainable agriculture example is replaced by a larger water-energy-food nexus: 40 concepts in five modules, with three ready-made scenarios.	Section 3.5
Safer import	Imported files and the model saved in the browser are checked and repaired where needed; the tool lists what it corrected. A problem in one tab no longer affects the rest of the tool.	Sections 3.3, 14.1
Scenario comparison	The baseline is always shown exactly once in the comparison chart and legend.	Section 9.6

Version 0.2 introduced modules and the analysis scope, the Influence Routes tab, the redesigned Network tab, and automatic saving in the browser. These are described throughout this manual.

1.3 How this manual is organized

Section 3 gets you started. Section 4 covers building a model on the Network and Model editor tabs, and Section 5 explains modules and the analysis scope, which apply to every tab. Sections 6 to 13 cover each analysis tab in turn, in the order they appear in the tool's own tab bar. Section 14 covers exporting your work, including the full structure of the exported Analysis Workbook. Section 15 is a glossary, and Section 16 lists known limitations and answers to common questions.

2. What Is a Fuzzy Cognitive Map?

A Fuzzy Cognitive Map (FCM) is a simple way of formalizing a causal mental model. It has just two kinds of ingredients:

- **Concepts:** the variables or factors that make up the system (e.g. "Fertiliser use", "Nutrient pollution"). Each concept has an "activation" value on a -1 (fully suppressed) to +1 (fully active) scale.
- **Relationships:** directed, signed, weighted causal links between two concepts (e.g. "Fertiliser use → Nutrient pollution, weight +0.75"). A positive weight means the source increases the target; a negative weight means it suppresses it. Weight strength runs from -1 (strongest suppressing) to +1 (strongest reinforcing).

Once a map like this is drawn, it stops being just a diagram and becomes a small dynamical system: you can give every concept a starting activation, let the relationships push those activations around for a number of iterations, and see where, or whether, the system settles into a stable state (an equilibrium). That is what the simulation-based tabs in this tool do in one form or another.

2.1 The update rule

At each iteration, every concept's new activation is computed from the activations of the concepts that point into it, passed through a squashing function that keeps every value on the -1..+1 scale. The Spaghetti Engine supports two variants of this update rule (configurable in Scenarios & Simulation → Advanced options):

Update rule	Formula	Meaning
Relative (default)	A(t+1) = squash(A(t) + W·A(t))	Each concept's new value builds on top of where it currently is. The formulation this tool uses by default.
Absolute	A(t+1) = squash(W·A(t))	Each concept's new value comes purely from incoming influence, discarding its own previous value entirely. This is the original Kosko FCM rule.

Where A(t) is the vector of every concept's activation at iteration t, and W is the matrix of relationship weights. The update repeats until the largest change in any concept's activation from one iteration to the next drops below a convergence threshold (it has "converged"), or a maximum number of iterations is reached. Without a squashing function the update is simply A(t+1) = A(t) + W·A(t) (relative) or A(t+1) = W·A(t) (absolute), and nothing keeps the values on the scale (Section 2.2).

2.2 Squashing functions

The squashing function maps a concept's total incoming influence back onto the -1..+1 scale. Three are available, and a fourth option removes the function altogether:

Function	Formula	Behavior
Hyperbolic tangent (default)	tanh(λ·x)	Smooth and symmetric; its slope at zero (its gain) equals λ.

Function	Formula	Behavior
Sigmoid	$2 / (1 + e^{-\lambda x}) - 1$	Also smooth and symmetric. It is exactly tanh(λx/2): the same curve at half the gain, so it responds only half as strongly to weak signals. The same weights can settle at a genuinely different (sometimes zero) equilibrium under sigmoid than under tanh at the same λ.
Trivalent (step)	+1 if x > 0.001, -1 if x < -0.001, else 0	A hard threshold with no smooth transition; every concept ends up fully on, fully off, or exactly neutral.
None (linear)	x	No squashing at all: the update is the plain linear one, and activation is neither bounded nor clamped, so the model's raw dynamics show. Feedback that reinforces itself makes activation grow without limit; such a run is stopped once any concept passes ±1000 and reported as diverging.

λ ("lambda", steepness) multiplies the input before it reaches the squashing function: higher values make the model behave more like a step function; lower values make it respond more gradually to the same influence. Because λ is the slope at zero, it acts as the loop gain of the whole system: below a critical value every run decays back to zero, above it activation can sustain itself. It has no effect on the trivalent function, which is already a hard step, and none without a squashing function.

> **Why this matters:** With every concept starting at exactly 0 and nothing locked or forced, tanh(0) = sigmoid(0) = 0, so the system cannot move away from zero on its own: it will converge back to 0 in a single step regardless of how the relationships are drawn. This is not a bug; it is the same reason a system sitting still at rest does not spontaneously start moving. Section 8.4 explains what to do about it.

Running without a squashing function. The None (linear) option is for exploring the real dynamics of the map. Without squashing, a map either decays towards zero (its feedback is weak or balancing enough), holds a constant level (some concepts are locked and everything else settles around them), or grows without limit (its feedback reinforces itself). It cannot saturate, swing between -1 and +1, or hide a runaway behind a bound. It is therefore a diagnostic: if a model only behaves in the bounded case, the squashing function is doing real work in it, and that is worth reporting (Annex, Section 3.3). Switch it on under Advanced options on the Scenarios & Simulation tab, where it applies to every simulation, or with the No squashing (linear) button on the Baseline Equilibrium tab, for that tab's run only.

> **What to expect:** With no squashing, even a small self-reinforcing loop grows by a constant factor on every pass. The tool stops such a run when any concept passes ±1000 (that value is only where the run happened to stop), marks it as diverging, and rescales the chart axes to fit. The Influence and Sensitivity scores on the Influence Metrics tab need a bounded scale, so they keep using the hyperbolic tangent while squashing is off.

2.3 Modules: a nexus as interacting subsystems

A nexus model describes several subsystems at once, for example water, energy, food, and the policies that govern them. In The Spaghetti Engine each concept can be given a **module** (its category), and concepts with the same module form one subsystem. Relationships then come in two kinds: those inside a module (how the subsystem works internally) and those between modules (how the subsystems are coupled, the nexus interface). Nothing

about any particular nexus is built in: modules are simply the categories you give your concepts, and the tool adapts to whatever you use. Section 5 explains how modules are used throughout the tool.

3. Getting Started

3.1 The interface at a glance

The Spaghetti Engine is a single page with a header bar, an analysis scope bar, and ten tabs arranged in four groups. A row of four group buttons sits above the tabs; click a group and its tabs appear beneath it. A fifth button, About & documentation, sits apart at the right (see the table below), and the tool opens on that page. The tool remembers which tab you last used in each group, and a link that takes you to another tab (for example from the Model editor to the Network tab) switches the group for you. The groups:

Group	Tabs and what they are for
Model building	Model editor, Network, Influence Metrics, and Influence Routes: building the model and reading its structure without running it. These are the tabs a participatory session mostly needs.
State and scenario analysis	Baseline Equilibrium and Scenarios & Simulation: where the model settles, and what changes under interventions.
Sensitivity analyses	Sensitivity Analysis and Monte Carlo: how much a result depends on one concept, and on the uncertainty in the weights. Mainly the analyst's tools.
Transition analyses	Transition Point Analysis and Transition Pathways: where, and in what order, interventions work best. Mainly the analyst's tools.
About & documentation	One page with a short description of the tool, who it is for, and why it was made; this manual, the Methodological Annex, and the Development Log as PDF downloads; and the contact details. It is not an analysis tab, so the analysis scope bar is not shown on it.

The ten tabs, in the order the tool shows them, are:

Tab	What it is for
Model editor	Bulk, table-style view of every concept and relationship. Also where you paste an adjacency matrix.
Network	The visual, drag-and-drop canvas with the Inspector. The primary way to build and edit a model.
Influence Metrics	Structural analysis (hubs, drivers, receivers, brokers) and the multi-scale comparison, with no simulation needed for the structural part.
Influence Routes	Chains of influence inside and between modules, the concepts they depend on, and the feedback loops in the system.
Baseline Equilibrium	Runs the model on its own, with no scenario applied, and shows where it settles.

Tab	What it is for
Scenarios & Simulation	Build "what if" scenarios (drivers, locks, overrides) and compare them against the baseline.
Sensitivity Analysis	Sweep one concept across a range and see how chosen outputs respond.
Monte Carlo	Re-run the model many times with the weights nudged at random, to see how far the results depend on the uncertainty behind the weights.
Transition Point Analysis	Sweep one intervention from 0 up to +1 (or down to -1) and find where its effect on an outcome changes fastest.
Transition Pathways	Combine interventions into ordered sequences, each implemented up to its transition point, and compare the pathways on gain, effort, and timing.

The header bar has these always-available controls:

- **Saved in this browser:** shows when the model was last saved automatically (Section 3.3).
- **New / example model:** starts a blank model, or loads one of three worked examples: the Water-energy-food nexus (40 concepts in five modules, Section 3.5), the Peat meadow nexus case study (Section 3.6), or the smaller Energy transition model. If a model is open, the tool first asks whether to replace it (Section 3.3).
- **Export Analysis Workbook:** builds and downloads one documented .xlsx file covering the model and every analysis you have run. See Section 14.3.
- **Export JSON / Import:** save the entire model (concepts, relationships, scenarios, settings) to a file, and load it back later or on another computer.

Directly under the tab bar, the **analysis scope bar** shows which part of the system you are looking at (for example "Viewing: Full Nexus, 40 of 40 concepts, 90 of 90 relationships") and lets you change it. It is described in Section 5.

3.2 Your first few minutes

1. Pick an example from New / example model in the header, or start a blank model and build your own.
2. Open the Network tab. On an empty map, double-click anywhere to add a concept; its name field is ready for typing. With nothing selected, the Inspector on the right shows a four-step guide to building a map.
3. Connect two concepts by dragging from one of the small dots on a concept's edge onto another concept, or select a concept and use Add a relationship in the Inspector. Then click the relationship to say whether it increases or decreases the other concept, and how strongly.
4. Give each concept a module in the Inspector, and keep an eye on the Model check below the map, which points out anything that still needs attention.
5. Open Influence Metrics to see the network's structure computed instantly with no simulation run, and Influence Routes to see how influence travels through it.
6. Open Baseline Equilibrium and click Run to see where the model settles on its own, then use Scenarios & Simulation to test interventions against that baseline.

3.3 Saving your work, and undoing mistakes

- **Automatic saving.** The model (concepts, relationships, scenarios, and settings) is saved in your browser a moment after every change. When you come back, a "Welcome back" notice says which model was restored and

offers to continue with it, start a blank model, or open the example.

- **What is not saved.** Analysis results are not saved; re-run the analyses you need after reopening the tool. The automatic save lives in one browser on one computer: clearing the browser's site data, using a private window, or switching computer means it is not there. Export JSON is the way to keep, share, or move a model.
- **Undo.** Ctrl+Z / Ctrl+Y step backward and forward through edits to concepts and relationships. After you delete a concept or relationship, a notice at the bottom of the screen says what was deleted and offers an Undo button.
- **Replacing a model.** Starting a blank model, loading an example, importing a file, or building from a pasted matrix replaces the current model. If a model is open, a dialog in the tool asks first, with three choices: Cancel, **Download a copy first, then replace** (saves the current model as a JSON file before replacing it), and **Replace model**. Press Escape or click outside the dialog to cancel. The replaced concepts and relationships can also be brought back with Undo until you reload the page.
- **Repairs on opening.** When the tool opens an imported file or the model saved in the browser, it checks it first. Problems that would otherwise break the display (concepts without a name, starting values or weights that are not numbers, two concepts with the same id, concepts without a position, relationships to concepts that do not exist, duplicated relationships) are corrected, and an amber notice lists exactly what was changed.
- **If something goes wrong.** A display problem in one tab stays in that tab: it shows "Something went wrong" with a Try again button, and every other tab and your model are unaffected. If the whole page fails, a recovery screen offers Reload, Download the saved model, and, as a last step, Start fresh.

3.4 Help inside the tool

- **Foldable explanation boxes.** Tabs open with a folded box ("What is sensitivity analysis?", "Purpose of this tab", "How these metrics are computed", and similar) that shows a one-line summary. Click the header row, or its Show button, to open it; the tool remembers whether you left each box open or closed.
- **Method & diagnostics panels.** Every calculation-based result is accompanied by a panel listing the key diagnostics of that run (settings used, number of simulation runs, caps reached) and, under "How this is calculated, and what it does not tell you", the method and its caveats in plain language.
- **Tooltips.** Hover over a column heading, a metric, or a control to see what it means.
- **Documentation and contact.** the About & documentation button (at the right of the group buttons, or the logo in the header) opens a page with this manual, the Methodological Annex, and the Development Log as PDF downloads, and the contact details of the developer (Section 3.1).

3.5 The example nexus model

The Water-energy-food nexus example shows how a larger model can be organised as sub-models. It is illustrative: its structure and weights demonstrate the tool, and it is not a calibrated model of any real system. It has 40 concepts in five modules of eight, and every concept has a definition:

Module	Concepts
Water	Freshwater availability, groundwater levels, total water demand, irrigation water use, water quality, drought frequency, wastewater treatment capacity, water price.
Energy	Energy demand, renewable energy capacity, fossil fuel use, energy price, bioenergy crop production, hydropower output, energy efficiency, grid flexibility and storage.

Module	Concepts
Food	Agricultural production, crop yields, fertiliser use, agricultural land area, livestock numbers, food security, food prices, food loss and waste.
Ecosystems & Climate	Greenhouse gas emissions, climate warming, biodiversity, soil health, natural habitat area, nutrient pollution, pollination services, carbon sequestration.
Governance & Society	Population and economic growth, household income, water allocation policy, climate and energy policy, agri-environmental measures, public environmental awareness, research and innovation, resource-use conflicts.

Of its 90 relationships, 50 lie inside a module and 40 couple the modules (water for crops and hydropower, energy prices for fertiliser, land for bioenergy, emissions for climate, policy levers from the governance module). Climate and energy policy and agri-environmental measures have no incoming relationships: they are policy levers that scenarios switch on. Without them, the baseline settles in a degraded state. The model comes with three scenarios, which all settle (converge) under the default settings:

Scenario	What it does
Climate and energy policy push	Holds climate and energy policy at full strength. Its effect spreads through every module.
Agri-environmental package	Holds agri-environmental measures at full strength. Its effect stays mostly in the Food, Ecosystems & Climate, and Water modules.
Integrated nexus policy package	Holds climate and energy policy, agri-environmental measures, and water allocation policy at full strength together.

Things to try: on the Network tab, choose Module overview to see how the five modules are linked, then open one module with Module view (Section 4.1); on Influence Routes, look at the routes and feedback loops between modules; on Scenarios & Simulation, tick compare on all three scenarios and run them to see which effects stay inside a module and which spread through the nexus; on Transition Pathways, add food security, biodiversity, and greenhouse gas emissions (lower is better) as outcomes and the policy concepts of the governance module as levers, and run the analysis to see which order of policies works best (Section 13).

3.6 The peat meadow example

The Peat meadow nexus (Dutch case study) is the illustrative case study of the transition pathways method: a food, climate, and biodiversity nexus in Dutch peat meadow polders. It has 20 concepts in five modules (Interventions, Water & soil, Farming system, Nexus outcomes, and Socio-economic) and 50 relationships, and every concept has a definition. Five interventions (raised groundwater levels, agri-environment payments, a nature-inclusive milk premium, area-based collective governance, and conversion to paludiculture) have no incoming relationships: they are levers that scenarios and pathways switch on. The three outcomes are food production, greenhouse gas emissions, and meadow bird biodiversity.

Two reinforcing feedback loops, the subsidence spiral (drainage, peat oxidation, and land subsidence lead to deeper drainage) and the intensification loop, settle the baseline into an almost saturated intensive state with high emissions and very low biodiversity. The example has seven scenarios: the baseline, each lever at full intensity, and a bundle of all five. Every lever responds slowly near its transition point, so this example loads with a limit of 300 iterations per run instead of 100 (Section 9.5); the Transition Point and Transition Pathways tabs run at least 1000 iterations in any case (Sections 12.6 and 13.2).

Things to try: run Baseline Equilibrium (it settles in 16 iterations); on Monte Carlo, run the baseline with weight uncertainty 0.1 and then 0.25, to see that the baseline is robust to small changes in the weights but not to large ones; switch off the squashing function (Section 2.2) to see the two loops run away; and on Transition Pathways, add the three outcomes (greenhouse gas emissions lower is better, biodiversity higher) and the five levers, and run the analysis.

4. Building a Model

4.1 The Network tab (recommended)

The Network tab is the primary way to build and edit a model: a canvas where concepts are cards and relationships are drawn as arrows, with the Inspector on the right and the Model check and module overview below. The map can be drawn in two styles, described first.

Figure and Editing style

The Figure / Editing switch in the toolbar chooses how the map is drawn. Figure style, the default, is meant to serve as a figure of the system:

- **Name-only cards.** Each card shows only the concept's name, in its module's colour. Hover a card to see its activation value.
- **Print-weight lines.** Lines are thin and still thicker the stronger the relationship. A decreasing relationship is dashed as well as orange, so the sign survives a greyscale print.
- **Lines that avoid cards.** Every line is straight, and bows only where it would run through another card or overlap the relationship in the opposite direction.
- **Labels that avoid each other.** Each weight is a small number on a white halo, placed along its line where it clears the cards and the other labels. Labels scale with the map when you zoom, like the rest of the figure.
- **A clean canvas.** The connection dots appear only when you hover a card or draw a relationship, and the background grid is hidden.

Editing style is the earlier look: larger cards that also show each concept's activation value, curved lines, and labels that keep the same size on screen at every zoom. The choice is remembered in this browser. It changes only how the map is drawn, never the model.

Adding and connecting concepts

- **Add a concept:** double-click empty space on the canvas, or use Add concept. The new concept is selected and its name field is ready: type the name and press Enter.
- **Connect concepts:** every concept has 12 small connection dots (three on each side). Drag from any dot and drop it on another concept to draw a relationship in that direction. The relationship attaches where you drew it from; otherwise it attaches wherever the line between the two concepts meets their edges. While you drag, the concept under the pointer is ringed and the connection dot the relationship would attach to lights up.
- **Without dragging:** select a concept and use Add a relationship in the Inspector: choose "affects" or "is affected by" and pick the other concept.
- **Re-attach a relationship:** select it, and two small crosshair handles appear at the two ends of its arrow. Drag either one onto a different concept; the far end stays where it is, the concept under the pointer is ringed, and

the connection dot the end would attach to lights up. Drop on empty canvas to cancel. A reconnection is one step in the undo history.

- **Move and arrange:** drag any card to reposition it. Layout never changes the model's behavior.
- When two concepts influence each other in both directions, the two arrows are automatically bowed apart so both remain readable.
- **Bend a relationship out of a tangle:** drag its label (or, with labels switched off, the small dot on the selected relationship) sideways, and the line bows out of the way and follows the pointer. Double-click the label or dot to straighten it. The bend is saved with the relationship. It is not offered on bundled links between folded modules, or in Orthogonal mode.

Reading a large model

Several tools keep a large map readable:

- **Hover to isolate.** Hover a concept to keep it and its direct connections at full strength and fade everything else; move away and the map returns to normal. The highlight clears a fraction of a second after the pointer leaves, so it does not flicker when the pointer crosses a label.
- **Find a concept.** The search box in the toolbar finds a concept by name. Choose a result, or press Enter for the first one, and the map selects the concept and centers on it; if its module is folded, it is unfolded.
- **Labels.** The Labels switch hides the numeric weight on every relationship, so each one is shown only by the color of its arrow (green increases, orange decreases) and its width (strength).
- **More room.** The map is as tall as the window allows. Collapse panel hides the Inspector so the map fills the width; Expand panel brings it back.
- **Arrows never cover concepts.** Relationship lines are drawn beneath the concept cards, so a concept can always be clicked or dragged, however many lines cross it.

The Inspector: editing a concept

Select a concept to see and edit it in the Inspector:

Field	What it does
Concept name	What the concept stands for. Press Enter to confirm.
Module	The subsystem the concept belongs to. Existing modules are offered as coloured one-click buttons; type a new name to create a new module (existing names are suggested as you type, which avoids accidental near-duplicates).
Definition	What the concept means and how it is measured. The same definition is shown and editable in the Model editor and is included in the exports.
Starting value	The concept's initial activation A(0), from -1 (very low) through 0 (neutral) to +1 (very high), set with a slider or typed exactly. Used as the starting point of simulations.
Relationships	Lists what the concept affects and what affects it, with each relationship's weight. Click one to open it. Add a relationship creates a new one from here.
Copy / Delete	Copy the concept (paste with Ctrl+V), or delete it together with its relationships (Undo is offered).

The Inspector: editing a relationship

Select a relationship to see it described in words, for example "When Rainfall increases, Crop yield decreases (a strong effect)." Below that:

Control	What it does
From / To	The two concepts; click either name to jump to that concept.
Effect	Increases (+): more of the source leads to more of the target. Decreases (-): more of the source leads to less of the target.
Strength	Weak (0.25), Moderate (0.5), Strong (0.75), or Very strong (1), keeping the chosen direction.
Fine-tune the weight	A slider, an exact number field, and nine labelled presets from "Very strong -" (-1) to "Very strong +" (+1).
Description	Optional: why the relationship exists (evidence, source, stakeholder).
Reverse direction	Makes the arrow point the other way (unavailable if a relationship in that direction already exists).
Delete relationship	Removes it (Undo is offered).

Auto Arrange

The Auto Arrange menu offers four smart layouts, designed for readable causal maps of 10 to 200 concepts, and five classic ones:

Layout	Best for
Readable (smart, the default)	Most models: the fewest crossings and a clean structure, with tightly linked concepts grouped together.
Systems thinking (smart)	Reading cause to effect: drivers (concepts that mainly influence others) on the left, outcomes (mainly influenced) on the right, and short feedback loops drawn as rings.
Cluster view (smart)	Comparing subsystems: clearly separated groups, by module when the model has modules, otherwise by the groups found in the relationships.
Centrality view (smart)	Finding the core: the most connected and influential concepts in the middle, peripheral ones towards the edge.
Hierarchical (top-down)	Models with a clear directional flow, laid out in layers.
Left-to-right causal chain	The same layered layout, rotated for a left-to-right causal read.
Circular	A quick, even overview of every concept with no implied hierarchy.
Force-directed	Physics-based layout that pulls connected concepts together and pushes everything else apart; good for spotting clusters.
Grouped by module	Models with modules: every module gets its own area of the map, so the modules read as separate sub-models.

A smart layout first finds groups of tightly linked concepts, places the groups, and then moves and swaps concepts to remove crossings, overlapping cards, lines running through cards, and labels that clash. The same model always gives the same arrangement. The cards glide to their new places, and a line under the toolbar reports the number of crossings, overlapping cards, groups, and feedback structures, and the time taken; hover it for the full layout score. A large model can take a few seconds, and the map stays usable meanwhile. Section 4.7 of the Methodological Annex describes the method. The classic layouts now also push apart any cards they leave overlapping.

Auto Arrange only assigns new positions; nothing about the model's structure or behavior changes, and you can keep dragging cards manually afterward. One undo (Ctrl+Z) restores the previous arrangement. When the analysis scope shows only part of the model, Auto Arrange arranges the visible concepts and leaves the positions of hidden ones untouched.

Other canvas tools

Control	Effect
Figure / Editing	Chooses the map style (Figure and Editing style, above).
Curved (Direct) / Orthogonal	Switches every relationship's line style. In Figure style the first option is called Direct: straight lines that bow only where they must.
Labels	Hides or shows the numeric weight on every relationship; with the labels hidden, each relationship is shown by its arrow's color and width only.
Find a concept	Searches the concepts by name and centers the map on the one you choose (Reading a large model, above).
Collapse panel / Expand panel	Hides or shows the Inspector, so the map can use the full width.
Hide / Focus	How concepts outside the analysis scope are shown: removed from the map (Hide) or kept but faded (Focus). See Section 5.4.
Export PNG	Downloads the map as an image. In Figure style: the whole map, cropped to its content, at three times screen resolution, without the zoom controls, minimap, or connection dots, whatever the current zoom. In Editing style: the map exactly as shown (layout, colours, zoom).
Undo / Redo (Ctrl+Z / Ctrl+Y)	Steps backward/forward through edits.
Ctrl+C / Ctrl+V	Copies and pastes the selected concept (with a small position offset).
Delete / Backspace	Deletes the selected concept or relationship (not while you are typing in a field).
Scroll / Shift+scroll	Zoom / pan horizontally.
Space + drag, or middle/right-drag	Pan the canvas.

Colour coding: Concept cards are coloured by module, and the legend under the map lists the module colours. Each module keeps its colour while other modules are added or renamed. In Editing style, the activation value on each card is coloured from red (negative) through grey (neutral) to blue (positive); before any simulation it shows the concept's starting value, after a run the simulated value, and the card says which one you are looking at. Relationships are green when positive and orange when negative, and thicker the stronger they are; in Figure style a negative relationship is also dashed, and the activation value appears when you hover a card.

Model check

Below the map, the Model check runs automatically while you build and lists anything worth attention, each with a short explanation. Click a name in the list to jump to that concept or relationship on the map; if it is outside the current analysis scope, the scope widens so you can see it.

Check	Why it matters
Add a second concept / Draw your first relationship	Next steps for a model that is just starting.
Concepts still called "New concept"	Unnamed concepts are hard to interpret in results and exports.
Names used twice	Two concepts with the same name are easy to confuse.
Concepts not connected	A concept without relationships cannot affect or be affected by anything, so it never moves in a simulation.
Relationships with weight 0	A weight of 0 means no effect, so the relationship does nothing.
Concepts without a module	When other concepts have a module, these are left out of module views and between-module analyses.
Module names that look alike	For example "Wter" and "Water": if it is a typo, the concepts are otherwise analysed as two separate subsystems.

Folding modules: the model at three levels

When a model has two or more modules, each module can be folded into a single block, so it works as a sub-model of the nexus. The structure switch above the map chooses what the map shows:

Choice	What the map shows
Full model	Every module unfolded and every concept shown.
Module overview	Every module folded into one block, with the links between the modules. This is the meta view of the nexus: which subsystems influence which, and how strongly.
Module view…	One module opened as its own sub-model, drawn inside an outline, with the other modules folded around it, so you see the module in the context of the rest of the nexus.

- **A folded module** is drawn as a stacked block in the module's colour, stating its number of concepts, its relationships inside, and its relationships to and from other modules. Double-click it, or click Unfold, to open it; **Own model** sets the analysis scope to that module alone (Section 5).

- **An opened module** is drawn inside an outline with a Fold button (to fold it again) and an Own model button.
- **Links between folded modules** bundle every relationship between the same two blocks into one arrow, labelled with the number of relationships and their net weight, for example "3 links · +0.50". The arrow is green when all of them increase the target, orange when all decrease it, and violet when they are mixed.
- **Select a folded module** to see, in the Inspector, its concepts (click one to unfold the module and select it) and its links with every other module (relationships out and in).
- **Select a bundled link** to see the relationships it contains, with their weights. Click one to unfold the modules involved and edit that relationship.
- **Dragging a folded block** moves every concept inside it by the same amount, so the module keeps its own layout when it is unfolded. In the module overview the blocks are drawn closer together than their concepts, so the overview fills the map.

> **Folding changes only the map:** Folding is a way of looking at the model. The model itself and every analysis are unaffected: analyses always use the analysis scope (Section 5), whether modules are folded or not. With fewer than two modules in view there is nothing to fold against, and the map always shows every concept.

Modules in this model

The "Modules in this model" box under the map explains what modules are, how the analysis scope and the Hide / Focus display use them, and summarises every module: its number of concepts, its internal relationships, how many relationships it sends to and receives from other modules, which modules it is directly connected to, and whether it is in the current scope. "Analyse on its own" sets the scope to that module.

4.2 The Model editor tab

The Model editor gives the same model as a bulk, table-style view, better for reviewing or editing many concepts and relationships at once than for freehand construction. The Model check (Section 4.1) is shown at the top.

- **Concepts panel:** every concept's name, module (with suggestions from the modules already in use), starting value, and **definition** (the box under each concept: what the concept means and how it is measured), editable inline. "show" opens the concept on the Network tab; the bin icon deletes it (Undo is offered).
- **Relationships panel:** every relationship's source, target, and weight (slider, number, and presets, as on the Network tab), with the plain-language sentence underneath, also addable and removable here.

Building from an adjacency matrix

The "Build from an adjacency matrix" panel lets you paste a square weight matrix directly out of Excel: concept names as both the column headers and the row labels, and cell [row, col] holding the weight of the relationship row → col. Clicking Build model from matrix parses it and, after confirmation, replaces the current model with the concepts and relationships it describes.

Exporting the model tables

The "Export model tables" box under the two panels exports the model itself (not the analysis results) as tables, for documentation, for a paper's supplementary material, or for use in other software. It always contains the complete model, whatever the analysis scope.

Button	Contents
All three as Excel (.xlsx)	One workbook with the three sheets below: Concepts, Relationships, and Adjacency Matrix.
Concepts with definitions (CSV)	ID, concept, module, definition, starting value, and number of relationships out and in, for every concept.
Relationships (CSV)	From, to, weight, effect (increases/decreases), strength (weak to very strong), the relationship in words, the modules at both ends, whether it runs between modules, its description, and the ids of both concepts.
Adjacency matrix (CSV)	The square weight matrix: concept names as row labels and column headers, cell [row, column] holding the weight of the relationship row → column, and 0 where there is none. It can be pasted back into Build from an adjacency matrix.

> **Number formats:** Both a plain decimal point ("0.5") and a European decimal comma ("0,5") are recognized. A comma that looks like thousands-grouping (e.g. "1,234") is treated as grouping, not a decimal point.

Guidance: setting starting values

The Model editor includes an in-app explanation (How to set each concept's "Initial" value) of what the starting value is for and how it should be used. In short: leaving every concept at 0 is a valid, deliberate structural-check setup, but it can never produce a non-zero equilibrium (Section 2.2); for a baseline that reflects an observed or elicited system state, give at least the currently active concepts a real starting value, and use the Scenarios & Simulation tab, not this field, for testing interventions. To try other starting values without changing the model, use the custom initial state on the Baseline Equilibrium tab (Section 8.2).

5. Modules and the Analysis Scope

5.1 What a module is

A module is a group of concepts that share the same category. Modules stand for the subsystems of the nexus you are modelling: sectors such as water, energy, and food, environmental domains, or policy arenas. Every distinct category you use becomes a module automatically; there is no fixed list.

- Set a concept's module in the Inspector (Section 4.1) or on the Model editor tab.
- Capitals and extra spaces are ignored, so "Water" and "water " are the same module. A module is labelled by the first spelling used.
- Concepts without a category form the Uncategorized module.
- Each module gets its own colour on the map, in the scope bar, and in module tables.

5.2 The analysis scope bar

The scope bar sits under the tab bar on every tab. It shows what you are viewing (for example "Viewing: Water Module", "Viewing: Food + Water", or "Viewing: Nexus Interface Mode"), how many concepts and relationships that includes, and the controls to change it:

Control	What it does
Analysis scope menu	Full Nexus; any single module ("Water Module"); Nexus Interface Mode; or Custom module selection, which points you to the module buttons.
Modules / Nexus Interface	Modules keeps every relationship among the selected modules. Nexus Interface keeps only relationships that cross from one module to another and hides those inside a module (needs at least two modules).
All, and the module buttons	All selects every module. Click a module button to add it to or remove it from the scope; "only" shows that module on its own. At least one module (two in Nexus Interface mode) always stays selected.
Show isolated concepts	Whether concepts left without any relationship inside the scope stay in.
Back to full nexus	Returns to the whole system.

When anything is left out, the scope bar turns amber, so a partial view can never be mistaken for the whole system.

5.3 What the scope changes

The scope applies to the whole tool, not just the map:

- **Every analysis uses only the network in scope:** the metrics on Influence Metrics, the routes and feedback loops on Influence Routes, the baseline equilibrium, every scenario, the sensitivity sweeps, the Monte Carlo runs, the transition point analysis, and the transition pathway analysis.
- **Each analysis tab says so.** While a partial scope is active, an amber note at the top of every analysis tab states which part of the system the results describe and how to read them.
- **Results from another scope are marked out of date.** Changing the scope after running an analysis shows the same "out of date" banner as editing the model; run it again to refresh.
- **New concepts join the module in view.** A concept added while a single module is in scope gets that module, so it does not disappear the moment it is created.
- **Exports follow the scope** (Section 14).

5.4 Hide and Focus on the Network tab

The Hide / Focus toggle on the Network tab decides how concepts and relationships outside the scope are drawn. **Hide** removes them from the map, and the map zooms to what is left. **Focus** keeps everything on the map but fades what is outside the scope, so a subsystem can be seen in the context of the whole. In both modes, faded or hidden elements are excluded from every analysis, and no concept ever changes position.

5.5 Reading results for part of the system

> **Subsystem results:** Analysing one module, or a combination of modules, analyses that subsystem in isolation. Relationships to concepts outside the scope are cut, so feedback that runs through another module is not represented. A concept's influence, sensitivity, or equilibrium can legitimately differ, sometimes a lot, from its value in the full nexus; that difference is the effect of cutting those links, which is exactly what subsystem analysis is meant to reveal.

Nexus Interface results: Nexus Interface Mode keeps only relationships between modules. Structural measures (degree, centrality, betweenness) then describe each concept's role in coupling the subsystems together. Simulation-based results describe a network made of those coupling relationships alone, which is a deliberately artificial model: use them to compare coupling structures, not as predictions of how the system behaves.

6. Influence Metrics Tab

This tab answers "which concepts drive the system, which absorb change, which sit at its center, and which connect its parts?" for the network in the current analysis scope.

6.1 How these metrics are computed

The foldable box at the top of the tab explains every metric in the table below. It distinguishes two kinds of metric that behave very differently underneath: structural metrics, computed directly from the relationships you have drawn, and perturbation-based scores, computed by running the model. This section summarizes the same explanation.

6.2 The full table

The Influence metrics table lists every concept in scope with all of the metrics below. **Click any column heading to rank the concepts by that measure**; click it again to reverse the order. The # column gives each concept's rank on the sorted measure, where #1 is always the highest value (reversing the order keeps the ranks), and concepts with exactly the same value share a rank. The sorted column is highlighted, and a caption above the table says what it is ranked by. A Module column appears when the model has more than one module. Click a row to select that concept (it is then selected on the Network tab).

Structural metrics, computed directly from the relationship weights with no simulation:

Metric	Definition
Indegree	Sum of the absolute weights of a concept's incoming relationships.
Outdegree	Sum of the absolute weights of a concept's outgoing relationships.
Net influence	Outdegree - Indegree (positive = more of a driver; negative = more of a receiver).
Centrality	Indegree + Outdegree.
Driver score	Outdegree ÷ Centrality: the share of a concept's connection weight that is outgoing (0 to 1).
Receiver score	Indegree ÷ Centrality: the share that is incoming. Driver and receiver scores add up to 1.
Betweenness	The share of shortest directed routes between every other pair of concepts that pass through this one (0 to 1). High values mark brokers that connect otherwise distant parts of the system.
Role	Hub, Driver, Receiver, Connector, or Isolated (see below).

Role is assigned from the degree figures:

Role	Condition
Isolated	Zero centrality; no relationships at all.
Hub	Top quartile of centrality among all concepts (≥ 75th percentile), and at least one relationship in each direction.
Driver	(Not already a Hub.) Outdegree more than 1.2× its indegree.
Receiver	(Not already a Hub.) Indegree more than 1.2× its outdegree.
Connector	Everything else; meaningfully connected both ways, without being exceptionally central.

6.3 Influence and Sensitivity scores (perturbation-based)

These two scores come from running the model. For each concept in turn, the model is set to all zeros, that one concept is locked at full activation (+1) for the whole run, and the simulation runs to equilibrium under the current squash function, λ, update rule, and convergence settings. Every other concept always starts at exactly 0 for this, regardless of the starting values set on the model, so these scores measure the network's structure in isolation.

- **Influence:** the average absolute movement a concept causes in every other concept once its own driven run settles.
- **Sensitivity:** the average absolute movement a concept undergoes, across all of the other concepts' individual driven runs.

A concept with high influence is one whose activation, if it changed, would ripple widely through the rest of the system. A concept with high sensitivity tends to move regardless of which other concept is driving the change. Computing these takes one full simulation run per concept in scope. With no squashing function selected, these scores use the hyperbolic tangent instead, because they average absolute responses and need a bounded scale.

6.4 Multi-scale analysis

The Multi-scale analysis section scores every concept at three fixed scales, whatever the current analysis scope: the **full nexus**, **its own module on its own**, and the **nexus interface** (every module, only relationships between modules). Each scale is a separate network, recomputed from scratch and ranked on its own, so a concept can be #1 inside its module and far down globally; that difference is the point.

- **Leverage points by scale:** the three concepts with the highest influence score at each scale (the full nexus, every module, and the nexus interface). With a custom scope, a card for the current scope is added.
- **Hierarchical ranking:** every concept's value and rank at each scale, for a chosen metric (influence score, centrality, betweenness, or out-degree). Click a column heading to sort by that scale's rank. With a custom scope, a column for the current scope is added.
- **Importance mainly from:** a reading aid naming the scale at which a concept ranks highest by percentile: system-wide position, within-module dynamics, or bridging between modules. It reports "Important at every scale" when a concept is in the top quarter at every scale and "Peripheral at every scale" when it is never in the top half. The margins behind this label are conventions of the tool; report the ranks themselves if the classification matters to an argument.

The multi-scale comparison needs a simulation run per concept at every scale, so on a large model it takes a moment; it is computed after the tab has opened and says "Computing multi-scale ranks" meanwhile.

6.5 System overview

Statistic	Meaning
Network size	Concepts and relationships in scope.
Density	Actual relationships ÷ maximum possible (n × (n-1)), as a percentage.
Connectivity (average degree)	(2 × relationships) ÷ concepts, with the maximum and minimum centrality.
Feedback structure	Simple feedback loops up to 8 relationships long (capped at 300 found), split into reinforcing and balancing. See Section 7.5 for the breakdown by module.
Complexity score	Receiver concepts ÷ Driver concepts (the Özesmi & Özesmi (2004) FCM complexity index). Higher = the model leans outcome-heavy; lower = driver-heavy.
Connectivity score	Relationships ÷ concepts; a simpler density measure, not normalized by the maximum possible.
Centralization score	How concentrated influence is in a few hub concepts versus spread evenly, 0 to 100%.
Clustering coefficient	Of the pairs of concepts linked to a concept, the share that are also linked to each other, averaged over concepts with at least two neighbours. High = tightly knit groups; low = chain- or star-like structure.
Modularity (Q)	How much more tightly the modules are connected internally than to each other, compared with chance. Clearly positive = the modules follow the network's structure; near zero = they do not; negative = they cut across it. Needs at least two modules in scope.
Interface relationships	Relationships between concepts of different modules, as a count and as a share of all relationships in scope.
Top broker	The concept with the highest betweenness.

6.6 Network metrics by category

When concepts have modules, this section aggregates every metric by module. Its table can be sorted by any column (click a heading), and includes the number of concepts, internal relationships, incoming and outgoing relationships between modules, the mean of each concept metric, and the internal density. Four charts compare the modules on mean influence, mean sensitivity, mean centrality, and connection density, each showing every module's name in full. An optional category interaction matrix shows, for every pair of modules, the number of relationships and their total signed weight. Concepts with no module are grouped under "Uncategorized" rather than dropped.

7. Influence Routes Tab

This tab traces how influence travels through the system. It works on the structure of the map alone and on the network in the current analysis scope. A foldable box at the top ("What is the Influence Routes tab?") explains it in the tool itself.

7.1 What an influence route is

An influence route is a chain of relationships along which a change in one concept can reach another: if A affects B and B affects C, then A → B → C is a route from A to C, even though A and C are not directly connected. Many of the most important nexus effects are indirect in exactly this way: a decision in one sector reaches another sector only through a sequence of intermediate concepts. Routes make those sequences visible, so you can see how the subsystems are coupled, which concepts act as bridges between them, and where coordination between sectors or policy domains would have to happen.

7.2 Route types

Route type	Question it answers	Which routes are included
Cross-module (Nexus Interface)	Where does influence leave one subsystem and enter another?	Every route that crosses at least one module boundary.
Between two modules	Through which concepts does module A affect module B, and B affect A?	Routes that start in one module and end in the other, through any concepts in scope; both directions or one.
Within one module	How is influence passed on inside this subsystem?	Routes that never leave the chosen module.
All routes	What are the strongest indirect effects anywhere in scope?	Every route in scope.

7.3 Steps

1. Set the analysis scope in the bar at the top (Section 5.2).
2. Choose a route type, and for "Between two modules" or "Within one module" the module(s) and direction.
3. Optionally narrow the selection with a From concept, a To concept, and the route length (from 1 to 6 relationships; the default is 2 to 4).
4. Read the bridge table and the transmission table first, then the list of individual routes. Click any concept to select it.

7.4 Reading the results

Result	What it shows
Strength	The product of the absolute weights along a route, so a route is never stronger than its weakest relationship and long chains fade.
Sign	Whether the first concept pushes the last one in the same (+) or the opposite (-) direction: the product of the signs along the route.
Bridge concepts & coupling points	For every concept a route passes through (not its start or end), the share of the selected routes' combined strength that passes through it, and the number of routes.

Result	What it shows
Boundary crossings	How often a route changes module at a concept: a coupling point between subsystems.
Bottleneck	A concept carrying at least half of the combined route strength: the selected routes largely run through it, which makes it a place where coordination between subsystems is needed.
Transmission routes between modules	Routes grouped by the module they start in and the module they end in, with their number, combined strength, and how many are positive and negative.
Strongest routes	The individual routes, strongest first, each concept coloured by its module. Export CSV downloads the full list.

Route enumeration stops at 6 relationships per route, 20,000 routes, and 250,000 search steps, so a dense network cannot freeze the page. If a selection reaches that limit, the tab says so and the counts and shares are based on the routes found before it; narrow the selection for complete figures.

7.5 Feedback loops

A feedback loop is a route that returns to where it started (A → B → C → A), so a change can amplify or dampen itself. Reinforcing loops (an even number of negative relationships around the loop) amplify change; balancing loops (an odd number) counteract it. The Feedback loops section at the bottom of the tab counts them:

- **Whole system:** every loop in scope, split into reinforcing and balancing. These are the same loops, with the same limits (up to 8 relationships long, at most 300 loops), as the loop count on Influence Metrics.
- **Within modules:** loops whose concepts all belong to one module, per module.
- **Between modules:** loops that run through two or more modules, grouped by the combination of modules they connect (e.g. Water ↔ Food ↔ Ecosystems). These are the feedback processes that couple the subsystems: a change in one module returns to it only after passing through another.
- **Per module:** for each module, the loops entirely inside it and all loops passing through it.
- **The loops themselves,** strongest first (strength is the product of the absolute weights around the loop), filterable to loops within a module, between modules, or involving a chosen module.

In Nexus Interface mode only relationships between modules are kept, so no loop can stay inside a module there.

7.6 What this tab is not

> **Structure, not simulation:** Routes and loops show where influence can travel given the relationships you drew, not how much actually arrives once feedback and the transfer function act on it. Use Scenarios & Simulation and Sensitivity Analysis for that. Nor are influence routes policy or transition pathways: they are causal routes through the map's structure.

8. Baseline Equilibrium Tab

This tab shows how the model behaves entirely on its own, before testing any intervention; it is the reference point every scenario on the Scenarios & Simulation tab gets compared against. It always runs with no scenario applied: no driver, lock, or override from anywhere else in the app affects it, and running it never affects anything else either. The foldable "Purpose of this tab" box explains the run and the role of the steepness λ.

8.1 Running it

The run follows the model's current update rule until it converges or hits the iteration cap (both set in Scenarios & Simulation → Advanced options). Two options next to the Run button apply to this tab only, without changing what any other tab uses:

- **Hyperbolic tangent / Sigmoid / No squashing (linear):** the squash function for this run. With No squashing (linear) the steepness control is hidden, and a short note explains that nothing then bounds the activation (Section 2.2).
- **Steepness λ:** a slider and number field for this run's λ, with "reset" to return to the shared setting. Because λ sets the loop gain of the whole model (Section 2.2), sweeping it here shows whether an equilibrium of zero is a property of the model or only of the chosen steepness.

8.2 The initial state vector

The Initial state vector box sets A(0), where the run starts:

- **Model's initial values:** each concept's starting value as set in the Inspector or on the Model editor tab.
- **Custom for this run:** set a starting value for each concept here with a slider or number field, or use Set all (model values, 0, +0.5, -0.5, +1, or random values between -1 and +1). Values that differ from the model's are marked "changed". These values are used by this tab only; Scenarios & Simulation and the other tabs keep using the model's values.
- **Save as the model's initial values** copies the custom values into the model itself, so every other tab starts from them too.

Running the baseline from several different starting vectors is the standard check of whether the model has one equilibrium (the same end state from every start) or several (the end state depends on where it starts).

8.3 Equilibrium diagnostics

Field	Meaning
Initial state type	"All zeros", "Model's initial values", or "Custom (this tab)": the starting state the displayed run actually used.
Converged	Whether the run settled below the convergence threshold before the iteration cap.
Iterations to convergence	How many update steps it took. A run that hits the cap shows "capped at N"; a runaway shows "stopped at N (runaway)".
Maximum residual change	The largest per-concept change between the final two iterations.
Fixed point detected	Mirrors Converged; in this engine a converged run is, by definition, a detected fixed point.

Field	Meaning
Stability	"Stable Equilibrium" (converged), "Oscillation" (settled into a short repeating cycle instead of a fixed point), or "Non-Convergence" (neither within the iteration cap). "Divergence" (only without a squashing function: activation grew without bound and the run was stopped, Section 2.2).

Below the diagnostics, a convergence plot shows every concept's activation at every iteration, and the iteration table can be exported as CSV. Customize & export opens the figure panel for the plot (Section 14.2); without a squashing function the vertical axis rescales to fit the values. The lines are drawn smooth, without a marker at every iteration; the marker size in the figure panel adds them.

8.4 If every concept starts at zero

As explained in Section 2.2, an all-zero start with nothing locked or forced can only ever converge straight back to zero, for any squash function this tool offers. This is a mathematical property of the update rule, not a bug. Before you run, the tab shows a notice about this, pointing to the initial state vector on this tab, the Model editor, and the driver controls on the Scenarios & Simulation tab, whichever you need.

8.5 If it looks out of date

If you edit the model (a concept's starting value, a relationship's weight, or an Advanced option) or change the analysis scope after running, an amber banner appears: "The model or the network view has changed since the baseline equilibrium was last run...". If only the initial state vector on this tab changed, a separate banner says the results still start from the previous vector. The numbers shown are exactly what they were at the time of the run; click Run again to refresh them. Moving a concept on the Network canvas does not trigger this, because position has no effect on the simulation.

9. Scenarios & Simulation Tab

9.1 What a scenario is

A scenario is a "what if" copy of your model: the same concepts and causal relationships, but with one or more interventions applied: holding a concept at a fixed value, overriding its starting value, or adjusting a relationship's weight. Running it re-computes where the system settles and compares that against the baseline. This lets you test a policy option, a shock, or a combination of measures inside the model before anything is implemented in reality. Scenarios run on the network in the current analysis scope. A foldable "What-If Scenario Analysis: how it works" box explains this in the tool.

> **Results are not a forecast:** A scenario shows what a specific causal structure, as you have defined it, implies under the assumptions you set, not a prediction of what will actually happen. Treat scenario output as a tool for reasoning about the system's structure and stakeholders' shared understanding of it.

9.2 Typical workflow

1. Make sure the baseline (no intervention) reflects a meaningful starting state; see Section 4.2's guidance on starting values.

2. Click New to create a scenario, or Duplicate an existing one as a starting point.
3. Give it a name and a short description of the policy or shock it represents (this description carries through to the exported workbook).
4. Apply your intervention(s): lock a concept's value, override a starting value, or adjust a relationship's weight.
5. Tick compare on the scenarios you want side-by-side, then run the simulation.
6. Read the comparison table and chart: check convergence, effect sizes, and whether concepts moved together (synergy) or in opposite directions (trade-off).

9.3 Setting up an intervention

For each concept, the editing panel offers:

Control	Effect
▲ / ▼ driver buttons	Forces the concept to full activation (+1 / -1) and holds it there for the entire run; the same convention Mental Modeler scenarios use.
Custom value field	Sets a specific starting value instead of full ±1, for finer-grained control.
Lock icon	Holds whatever value is set (custom or default) fixed for the entire run, independent of the driver buttons.

9.4 Glossary used on this tab

Term	Definition
Scenario	A named, self-contained variant of the model in which one or more interventions are applied, run and compared against the baseline without altering the baseline itself.
Intervention	A specific change applied within a scenario: a driver/lock, an overridden starting value, or an adjusted relationship weight.
Equilibrium	The state a scenario's concepts settle into once the model stops changing, given its interventions and the underlying causal structure.
Effect Size	How far a concept's equilibrium value in a scenario differs from its value in the baseline.
Synergy	When two or more interventions, run together, produce a larger effect than the sum of their individual effects run separately.
Trade-Off	When an intervention that improves one outcome simultaneously worsens another.

9.5 Advanced options

These settings, found under Advanced options on this tab, govern every simulation in the app (except where the Baseline Equilibrium tab overrides them for its own run), and are recorded in the Export Analysis Workbook.

Setting	Default	What it controls
Max iterations	100	The run stops after this many iterations even if it has not converged.

Setting	Default	What it controls
Convergence threshold	0.001	The run is considered converged once the largest per-concept change between iterations drops below this.
Update mode	Synchronous	Synchronous updates every concept at once from the previous iteration's values (reproducible). Asynchronous updates concepts one at a time in a randomly shuffled order each iteration, so results can differ between runs of the identical model.
Update rule	Relative	See Section 2.1; Relative vs. Absolute.
Squashing function	Hyperbolic tangent	See Section 2.2. Also offers None (linear, unbounded), which then applies to every tab that simulates.
λ (steepness)	1	See Section 2.2. Hidden when the squash function is Trivalent. Also hidden when there is no squashing function.

9.6 Reading the comparison

Once you run, this tab shows a bar chart and table comparing every compared scenario's final value for each concept against the baseline. The baseline is always part of the comparison and appears exactly once, labelled Baseline; its row has no compare box. Scenarios that share a name are numbered so every series can be told apart, and a note lists any ticked scenario that has not been run yet. For a single compared scenario it adds a trajectory comparison (baseline, scenario, and their difference, iteration by iteration) and a "Component change vs. baseline" chart expressing each concept's change in percentage points of the full -100% to +100% activation range, with every concept name on one line. That axis expands beyond ±100 if a concept's change genuinely exceeds it, e.g. flipping from fully negative to fully positive. Results can be exported as CSV. Without a squashing function the chart axes rescale to the values, and the method panel counts the runs that diverged. Customize & export opens the figure panel for the comparison chart and the trajectory charts (Section 14.2); the trajectories are drawn as smooth lines without markers, which the marker size in the panel can add.

9.7 If it looks out of date

The same amber banner described in Section 8.5 appears here too, and is a little broader: because a comparison also depends on each scenario's own drivers, locks, and overrides, editing a scenario's intervention, not just the underlying model or the scope, also marks the comparison out of date. Re-run the simulation to refresh it.

10. Sensitivity Analysis Tab

Sensitivity analysis evaluates how changes in one concept (the input) affect one or more other concepts (the outputs) elsewhere in the system. It works by holding an input concept at a series of fixed values across a range, re-running the simulation at each value, and recording where each output settles. It helps identify critical leverage points, robust variables, vulnerable variables, and high-impact interventions. The foldable "What is sensitivity analysis?" box explains it in the tool.

A highly sensitive concept changes significantly when system inputs are modified. A highly influential concept produces large changes in other concepts. The input concept is locked at each swept value for the whole run (the

same "driver" semantics used everywhere else in the app), so a held level, not just a starting point, is what gets tested. The sweep uses the network in the current analysis scope.

10.1 Steps

1. Choose the input concept to sweep.
2. Choose one or more output concepts to observe as the input changes (the input concept itself is excluded).
3. Set the range the input sweeps across, from Range min to Range max.
4. Choose how many increments to sample across that range.
5. Optionally, tick one or more scenarios under "Compare across scenarios" to run the sweep against those instead of just the active scenario.
6. Click Run analysis to generate the sensitivity curves, tornado chart, and elasticity table.

10.2 Reading the results

Output	What it shows
Sensitivity curves	Each output's equilibrium value plotted across the swept input range.
Tornado chart	Every other concept, swept across the same range one at a time, ranked by how much it moves the primary output: which concept the output is most sensitive to, not just the one you chose as input. It spans the full width of the page so every concept name fits on one line.
Impact ranking	The tornado ranking as a table (Rank, Concept, Impact) with the sign color-coded, scrollable when the model is large. Export CSV.
Elasticity metrics	One number per output: (output at max input - output at min input) ÷ (max input - min input), the average slope across the full swept range. Export CSV.
Response table	The raw input → output values behind the curves, for every sampled increment.

While no squashing function is selected, an amber note reminds you that a run that runs away is stopped where it happens to be, so curves and rankings that include such runs say little about the model.

11. Monte Carlo Tab

Every relationship weight in a map is a judgment: elicited from a stakeholder, taken from the literature, or estimated. The Monte Carlo tab asks how much that uncertainty matters. It re-runs the model many times, each time nudging every weight by a small random amount, records where every concept settles each time, and shows how widely the results spread. A concept whose result barely moves is robust to the uncertainty in the weights. One whose result swings widely is sensitive to it, and its nominal value should not be reported without the range. The foldable "What is Monte Carlo simulation?" box explains it in the tool.

This asks a different question from the Sensitivity Analysis tab. Sensitivity Analysis moves one concept deliberately, one at a time, to find leverage points. Monte Carlo moves every weight at once, at random, the way real uncertainty in elicited weights behaves. It is a check on how far to trust a result, not a search for where to intervene.

11.1 Steps

1. Choose the scenario to test (Baseline by default). The scenario's own overrides are respected: a relationship weight or a starting value that the scenario pins to a specific number is never nudged, and a locked concept stays locked.
2. Choose the number of runs: 100, 250, 500 (the default), 1,000, or 2,000. More runs give smoother statistics; the standard error of a mean falls with the square root of the number of runs.
3. Set the weight uncertainty (±): how far each unlocked weight may be nudged, up or down, on the same -1 to +1 scale as the weight itself. The default is 0.1. Every weight gets its own random amount, drawn uniformly, and stays within -1 to +1.
4. Optionally set the starting-value uncertainty (±) to nudge every concept's starting value as well. The default, 0, varies the weights only.
5. Note or change the random seed. The same model, settings, and seed always give the same result; the dice button draws a new seed.
6. Click Run simulation. A progress bar shows how far it is, and Cancel stops it. Leaving the tab also stops a run in progress.

11.2 Reading the results

A header states the scenario, how many runs settled, how long the run took, and the settings used (weight and starting-value spread, seed, transfer function, and update mode), so a result can be reported and repeated. The summary table has one row per concept; click a column heading to sort. It starts sorted by standard deviation, largest first, so the concepts most sensitive to the weights come first.

Column	Meaning
Nominal	The result with every weight exactly as recorded: the ordinary equilibrium of the scenario.
Mean / Median	The average and the middle of the results over all runs. A mean far from the nominal value shows that the recorded weights sit away from the center of what the noise produces, often because a concept saturates near +1 or -1.
Std dev	How much the result varies across runs. Concepts that nothing nudged (levers with no incoming relationships, and locked concepts) show zero.
P5 / P95	The 5th and 95th percentiles: only 5% of the runs landed below the P5 value, and only 5% above the P95 value. Together they give a plausible range that assumes nothing about the shape of the distribution.
Min / Max	The extremes over all runs.

Below the table, the Distributions box shows a histogram of the sampled outcomes for a concept. The three concepts with the largest spread are charted when a run finishes; click a concept to add or remove its chart. A histogram with two separate peaks often means that the model has more than one equilibrium and that the weights decide which one a run reaches. Export CSV downloads the summary table.

11.3 Runs that do not settle

Every run is simulated with the shared Advanced options (Section 9.5), and the header says how many settled. A banner appears when some did not: the runs that reached the iteration limit are still included in the table, but treat their values with extra caution. In a model that settles slowly, or lies close to an oscillation, small changes in the weights can push a run beyond the limit or into a cycle even though the nominal run settles; raising Max iterations helps only in part (Section 16.2). Without a squashing function, runs that grow without limit are stopped once a concept passes ±1000 and are counted separately; their values are only the stopping point, so the mean and spread of the concepts they affect are not meaningful.

11.4 Repeating a run

The seed makes a run exactly repeatable, including with asynchronous updating, whose random update order is drawn from the seed too. Report the seed, the number of runs, and the spreads with any Monte Carlo result. Different seeds give slightly different summaries; if a conclusion changes with the seed, use more runs.

11.5 If it looks out of date

As on the other analysis tabs, an amber banner appears when the model, the analysis scope, or a simulation setting changes after a run. The results shown are exactly those of the run; click Run simulation again to refresh them. Monte Carlo results are not saved, and they are not part of the Export Analysis Workbook.

> **What this does not tell you:** The noise is uniform and independent for every weight, with one spread for all of them, and it is applied to weights and starting values only: the structure of the map is taken as given. The spread is a stated assumption about how uncertain the weights are, not an estimate of it, so the result answers "what if every weight could be off by this much?", not "how likely is this outcome?". As everywhere in the tool, results are implications of the map as drawn, not forecasts.

12. Transition Point Analysis Tab

Transition Point Analysis identifies the intervention intensity from which additional effort adds markedly less to an outcome than the effort before it: the point of diminishing returns. It works by sweeping one intervention concept away from 0 in small steps, up to +1 (an increase) or down to -1 (a decrease), re-running the simulation at each step, and recording how each chosen outcome concept responds along the way. The foldable "What is a Transition Point?" box explains it in the tool.

The intensity on the horizontal axis always runs from 0 to 1 and means how far the intervention is pushed: for a decrease, an intensity of 0.4 means that the intervention is held at -0.4. The two directions are the two ways a scenario can drive a concept (a driver at +1 or at -1 on the Scenarios & Simulation tab), so the last step of each sweep is exactly that scenario, and the curve shows what happens on the way there. Effects are measured from the intervention held at 0, not from the baseline in which it is free to move. For a decrease, a positive slope means that the outcome rises as the intervention is pushed further down.

The marginal benefit of each step is how much the outcome moves over that step, per unit of intensity, counted in the direction the outcome moves over the whole sweep. The transition point for an outcome is the intensity at which this marginal benefit declines most steeply: up to there, extra effort keeps paying off; beyond it, each further step adds markedly less. If the marginal benefit does not decline before full intensity, the transition point is 1. The tool has no built-in notion of which direction is "good" for a given concept, which is why it counts benefit in the

direction the outcome moves: always check the sign of the effect in the results table before treating a transition point as a leverage point worth pursuing. The Transition Pathways tab (Section 13) does know which direction is better, because you state it for every outcome, and uses that to combine transition points into pathways.

12.1 Steps

1. Choose the intervention variable to sweep.
2. Choose its direction: increase from 0 to +1, or decrease from 0 to -1.
3. Choose one or more outcome variables to observe (the intervention itself is excluded), or use "All variables" to select every other concept at once.
4. Choose the intensity resolution (how many steps to sample across the sweep) and, optionally, adjust the flat-response threshold used to classify an outcome with little to no response.
5. Click Run analysis to generate the response curves, marginal-benefit curves, transition point table, and rankings.

12.2 Response type classification

Each outcome's response is classified by where its transition point falls, checked in this order:

Type	Condition
Flat	The outcome's total range across the whole sweep is below the flat-response threshold: no meaningful response, wherever the marginal benefit happens to decline.
Proportional	Transition point 1: the marginal benefit does not decline before full intensity (or declines most steeply only at the last step), so the benefit keeps accruing in proportion to the effort, or faster, over the whole range.
Early	Transition point at most 1/3: diminishing returns set in early, and most of the benefit comes at low intensity.
Middle	Transition point between 1/3 and 2/3: diminishing returns set in at an intermediate intensity.
Late	Transition point of 2/3 or more, but below 1: the benefit keeps building until high intensity before further effort pays off markedly less.

12.3 Synergy classification

When two or more outcomes are selected, their effects at full intervention intensity are jointly classified: Trade-Off (some improve, some worsen), Full / Partial / Narrow Synergy (a majority-to-all improve), Full / Partial / Narrow Negative Synergy (a majority-to-all worsen), or No Synergy (nothing moved beyond the flat-response threshold).

12.4 KPI cards and Efficiency

KPI	Meaning
Earliest / Latest transition point	The outcome whose returns diminish soonest / latest in the sweep. Flat outcomes are left out, unless every outcome is flat.
Highest total effect	The outcome with the largest overall change from start to end of the sweep.

KPI	Meaning
Most efficient lever	The outcome with the largest Efficiency score.

Efficiency = Total Effect ÷ Transition Point. It measures how much an outcome changed overall, relative to the intensity from which further effort pays off markedly less: a high efficiency means a small nudge produces a big change; a low one means most of the intervention's range is spent for comparatively little. Its sign matches the effect; a negative efficiency means the outcome got worse, not better.

12.5 The results table and figures

The Transition point results table lists every intervention and outcome combination that has been run, with the columns Intervention, Outcome, TP (the transition point), Change at TP (how far the outcome has moved by the transition point: its value at the TP minus its value at intensity 0), Share at TP (Change at TP as a percentage of Max Change; above 100% when the outcome overshoots its final value, and n/a when Max Change is below the flat-response threshold), Max Change (the effect size over the sweep), Range (the total movement of the outcome across the whole sweep; below the flat-response threshold it counts as no meaningful response), Efficiency (Section 12.4), and Response Type (Section 12.2).

Click any column heading to sort by it, and click again to reverse the order; an arrow shows the sorted column and its direction. Numeric columns sort by number, text columns alphabetically, and the sort stays in place when you change the outcomes shown, so the earliest, latest, or strongest transition points can be found without exporting the table.

The response curves and the marginal-benefit curves each have a Customize & export button that opens the figure panel (Section 14.2). Which outcomes are shown, and in what order (as selected, alphabetical, by transition point, or by effect size), can be set there, together with colors, fonts, and size; the figure is exported as PNG, SVG, or PDF.

Dashed vertical lines mark each outcome's transition point on both charts. The marginal-benefit chart plots each outcome's marginal benefit in the direction that outcome moves (a falling outcome is plotted as a positive benefit of falling), so the transition point is where its curve falls most steeply.

While no squashing function is selected, an amber note at the top of the tab reminds you that runs that run away are stopped where they happen to be, so curves that include them say little about the model.

12.6 Precision and convergence

The transition point compares neighbouring changes of the response curve, which magnifies whatever is left of a run that has not quite settled. Every run of this tab therefore continues until no concept changes by more than 1e-9 per iteration (or the convergence threshold of Advanced options, if that is smaller), with at least 1000 iterations (or the global limit, if that is higher). Whether a run counts as converged still follows the global threshold. Method & diagnostics shows how many runs converged and how many settled to 1e-9. If runs did not converge, the model keeps oscillating there, and the curves and transition points are not reliable: check the Baseline Equilibrium tab, or try the absolute update rule. Most models settle within a few hundred iterations, and a sweep takes a fraction of a second; a model that keeps oscillating runs to the limit at every step, which takes a few seconds at 500 increments.

A transition point can only fall on a sampled intensity, so its precision is one step (±0.01 at 100 increments). Re-run at a higher resolution before quoting a value. A marginal benefit that changes by less than 1% of its largest value per unit of intensity counts as not declining, so rounding noise on a straight response does not create a transition point.

13. Transition Pathways Tab

Transition Point Analysis asks at what intensity a single intervention is most effective. The Transition Pathways tab asks in what order interventions should be implemented. It combines intervention levers into **transition pathways**: ordered sequences of interventions that move the system, one stage at a time, from its Business-as-Usual state towards a desired state. Because every lever is tested in the state the earlier stages have created, the tab shows how interventions make each other more or less effective, which a comparison of levers one at a time cannot show. The foldable "What is Transition Pathway Analysis?" box explains the idea in the tool.

13.1 How a pathway is built

Every pathway starts from **Business-as-Usual (BAU)**: the equilibrium the model reaches from its own starting values with no scenario applied. It then proceeds in stages:

1. The first lever is swept from 0 to 1 in small steps, starting from BAU, and the tool finds its **transition point**: the intensity from which further effort adds markedly less to the improvement of the nexus outcomes, the point of diminishing returns of the total outcome gain. If the gain there is no higher than in BAU, the lever is not added.
2. The lever is implemented up to that transition point, and the system settles to a new equilibrium.
3. The lever stays in place. The next lever is swept from the new equilibrium, and its transition point is found again, now in the state the first stage has created: its **conditional transition point**.
4. This repeats for every further stage, up to the maximum number of stages.

If earlier stages make a lever's returns diminish sooner, its conditional transition point is lower than on its own: less of the lever is needed for most of its benefit, and the earlier stages **enable** it. If they push it later, more of it is needed, and they **delay** it. The difference is shown as the lever's **TP shift**. A stage is an implementation phase, not a period of time: the tool assumes each stage lasts long enough for the system to settle before the next one starts, and that implemented levers stay in place.

13.2 Setting up the analysis

The setup panel has four parts. The setup is remembered in this browser, so it survives a reload; concepts you delete from the model are removed from it automatically.

Part	What to set
1. Nexus outcomes	The concepts a pathway should improve, for example food security, biodiversity, and greenhouse gas emissions. For each: **Improvement means** Higher is better or Lower is better; a **Weight** (1 for every outcome weighs them equally); and, optionally, a **Desired level** on the -1 to +1 scale, after which each pathway reports how many desired levels it reaches.
2. Intervention levers	The interventions that may be combined, in any order. For each: **Intensity means Increase (0 to 1) or Decrease (0 to -1), as on the Transition Point tab: an intensity of 0.4 holds the lever at +0.4 or at -0.4, and full intensity is the same as setting the lever as a driver, up or down, in Scenarios & Simulation;** and an **Effort weight**, how demanding one unit of this lever is compared with the others (1 = the same). No cost data are needed: a ranking is enough. **Suggest levers** adds every concept that acts on others but is not acted on by anything, the usual shape of a policy instrument; review the suggestions before running.

Part	What to set
3. How pathways are built	The options in the table below.
4. Order constraints (optional)	Rules such as "lever A can only come after lever B", for a legal or practical prerequisite. Pathways that break a rule are not built.

Option	Default	Meaning
Maximum number of stages	3	The longest pathway, from 1 to 5 stages. Every shorter opening of a pathway is itself also evaluated as a pathway.
Stage intensity	Up to the conditional transition point	Each stage stops at the lever's transition point, as the method intends. **Full implementation (intensity 1)** puts every lever at full strength instead, as a reference case.
Allow scale-up stages	Off	With the transition-point rule: a lever implemented to its transition point may return once in a later stage and be raised to full intensity.
Pathway construction	Every order (exhaustive)	**Every order tries all orderings of the levers up to the maximum length, which is exact but grows fast (at most 20,000 pathways). Beam search keeps only the best few partial pathways at each stage (by outcome gain or by efficiency; 10 by default). Greedy takes, at each stage, the lever with the highest marginal benefit (the steepest step of its sweep).**
Intensity resolution	20 steps (±0.05)	How finely each lever is swept. A transition point can only fall on one of these steps, so this sets its precision (10, 20, or 50 steps).
Flat-response threshold	0.015	An outcome change smaller than this counts as no change: for the synergy score, for trade-offs, and for leaving out levers that move nothing.
Minimum gain (% of best)	25	A pathway is compared in the archetypes only if its total outcome gain reaches this share of the best pathway's gain (Section 13.5).
Maximum iterations per run	1000	Every stage sits at a transition point, where the model settles slowly, and transition points need runs settled to 1e-9, so pathway runs allow more iterations than the global setting; the limit is never below 1000. Whether a run counts as converged follows the global threshold. All other simulation settings are the global ones (Scenarios & Simulation, Advanced options); pathway runs are always synchronous.
Also test robustness	Off	Repeats the whole analysis with steepness halved and doubled, with the other squash function, and with the other update rule, and reports how often the top pathways keep their archetype and rank. About five times longer.

Below the setup, the tool estimates how many pathways and simulations the analysis will take. Click **Run pathway analysis**; a progress bar shows how far it is, and **Cancel** stops it. If every order would mean more than 20,000 pathways, the run button stays disabled and the tool suggests beam search or fewer stages. The analysis uses the network in the current analysis scope; outcomes and levers outside the scope are left out, with a note saying so.

13.3 Which levers take part

The analysis first tests every lever on its own, from BAU. This produces the **lever profiles** (Section 13.6). A lever is left out of the pathways if it moves no outcome by more than the flat-response threshold, or if the total outcome gain at its transition point is no higher than in BAU. Levers that improve some outcomes and worsen others are kept, because a later stage may make up for the trade-off.

> **The same definition as on the Transition Point tab, on a different curve:** The transition point on this tab is the marginal-benefit transition point of the total gain in the nexus outcomes, counted in the direction you marked as better, rather than of one outcome in the direction it happens to move. A stage stops at it, and a lever is only added if the gain there is higher than the gain the pathway has already reached.

A stage at full intensity (with Full implementation, or a scale-up stage) can lower the total gain, when the lever worsens the outcomes beyond its transition point. Such stages are kept, and their stage gain is shown in orange; the efficient frontier and the rankings show whether they are worth taking.

13.4 Pathway metrics

Metric	Meaning
TOG: Total Outcome Gain	How much the pathway improves the nexus outcomes in its final state, compared with BAU, each outcome counted in its desirable direction and weighted.
CII: Cumulative Intervention Intensity	The total effort: the sum of the final lever intensities, each times its effort weight. With effort weights of 1, a CII of 1 means one lever at full intensity.
PEI: Pathway Efficiency Index	TOG ÷ CII: outcome gain per unit of effort.
Avg TP: Average Transition Point	The mean conditional transition point of the stages. Low values mean that the levers' returns diminish early in their range: most of their benefit comes at low intensity.
SS: Synergy Score	How many nexus outcomes the pathway improves by more than the flat-response threshold. SS equal to the number of outcomes means every outcome improves.
EBI: Early Benefit Index	The area under the benefit accrual curve (share of the final gain realized against share of the effort spent). Above 0.5, benefits arrive ahead of the effort; below 0.5, they arrive late.
IG: Interaction Gain	The pathway's TOG minus the sum of what each of its levers achieves on its own from BAU at the same intensity. Positive: the levers reinforce each other. Strongly negative: each lever alone already moves much of the system, so their separate effects overlap.

Metric	Meaning
TP shift	The mean difference between the levers' conditional and stand-alone transition points. Negative: earlier stages make the levers' returns diminish sooner.

13.5 Pathway archetypes

Pathways are sorted into four archetypes, each answering a different strategic question. Only **admissible** pathways take part: those that converged at every stage, have no trade-offs (no outcome worse than BAU by more than the flat-response threshold), and reach the minimum gain. Thresholds such as "the lowest quarter" are taken across the admissible pathways of the same run, so an archetype describes a pathway compared with the other options, not an absolute standard.

Archetype	Rule	Ranked by
Low-Hanging Fruit	Effort (CII) in the lowest quarter, average TP at most 1/3, and EBI above 0.5. Quick wins under resource constraints.	Lowest effort first
High-Impact	TOG in the top fifth, and every nexus outcome improved. The largest overall improvement, whatever the effort.	Highest TOG first
Efficiency	PEI in the top fifth, at moderate effort (between the lowest and the highest quarter). The greatest gain per unit of effort.	Highest PEI first
Deep Investment	Effort in the highest quarter, average TP above 1/2, and EBI below 0.5. Sustained effort before substantial gains.	Highest TOG first

A pathway can belong to more than one archetype; its **primary archetype**, used for its colour on the chart, is the one in which it ranks highest. Ties in a ranking are broken by the synergy score, then the EBI, then the shorter pathway. An admissible pathway that meets no rule is shown as **Intermediate**; pathways with trade-offs as **Conditional (trade-off)**; the rest as **Below minimum gain** or **Did not converge**.

An archetype can be empty. Its card then says which of its conditions how many pathways meet, so you can see which condition nobody meets. In a model where one lever already moves the whole system, for example, the most efficient pathways all need very little effort and are Low-Hanging Fruit, and Efficiency stays empty. This describes the model, not a failure of the analysis.

13.6 Reading the results

Part of the results	What it shows
Method & diagnostics	Pathways evaluated, how many are admissible, simulation runs, pathways that did not converge, how many runs settled to 1e-9, the precision of the transition points, the simulation settings used, and how many lever sets were found to be path dependent. Below it, how the analysis works and what it does not tell you.
Archetype cards	The number of pathways in each archetype and its best pathway. Click the pathway to open it.

Part of the results	What it shows
Effort-gain plane	Every pathway by the effort it needs (CII, horizontal) and what it achieves (TOG, vertical), coloured by primary archetype. The dashed line is the **efficient frontier**: the pathways for which no other pathway achieves more with less. The vertical axis is zoomed to the data, because pathway gains can differ only slightly. Click a point to open that pathway. The selected pathway is labelled on the chart. Customize & export opens the figure panel, where the number of background points can be capped (the selected pathway, the efficient frontier, and the archetype pathways are always shown).
Rankings by archetype	The five best pathways of each archetype, with the value they are ranked by.
Selected pathway	The pathway's stages, archetypes, and eight metrics; a stage table (transition point alone and here, the shift, the intensity, the stage gain, the total gain after the stage, the stage's efficiency, and the value of every outcome at the end of the stage); an outcome trajectory chart (improvement of every outcome after each stage); and the benefit accrual curve behind the EBI. The outcome trajectory and the benefit accrual chart (a filled area whose shaded region is the Early Benefit Index) each have Customize & export.
All pathways	A sortable table of every pathway with all metrics, filtered by All, Admissible, Efficient frontier, one archetype, Conditional, or Not admissible. It shows 50 rows at a time; click a row to open the pathway.
Implementation portfolios	Every opening move, and the two-stage openings shared by the most top-ranked pathways. **Regret** is how much total outcome gain an opening gives up compared with the best pathway of all (0: the best outcome stays within reach). **Top pathways opened** counts how many of the five best pathways of each archetype start this way. A zero-regret opening that opens many top pathways is a **no-regret first move**. For each archetype, the table also shows the best way to continue.
Lever profiles	Each lever on its own from BAU: its transition point, response type, outcome gain at full intensity, synergy classification, its effect on every outcome with that outcome's own transition point, and whether it takes part in the pathways (and if not, why).
Invariance test	For the lever sets of the top pathways, every order of the same levers at the same final intensities is simulated. If the end states differ by more than 0.1, the order decides where the system ends up: the model is **path dependent** for these levers. Smaller differences come from where each run stops settling.
Robustness (when switched on)	For every top pathway, in how many of the four alternative settings it stays admissible, keeps its archetype, and stays among the five best of that archetype. A pathway that holds up across settings reflects the causal structure rather than a modelling choice.

The values of the outcomes at the end of each stage are reference levels for monitoring: in practice, the next stage would start once the outcomes have reached roughly these levels.

13.7 Exports

The Export card at the bottom of the results downloads all pathways, the stages of the selected pathway, the lever profiles, and the portfolios as CSV files, and the effort-gain plane, the selected pathway's outcome trajectory, and

its benefit accrual chart as figures (PNG, SVG, or PDF, from the figure panel, Section 14.2). The Export Analysis Workbook in the header also includes the pathway results (Section 14.3).

13.8 If it looks out of date

If you edit the model, change the analysis scope, or change the simulation settings after running the analysis, the results show the amber out-of-date banner. If you only change the setup, a grey note says that the results still describe the previous setup. In both cases, run the analysis again to update them.

> **What this tab does not tell you:** Pathways are implications of the causal structure as drawn, compared with each other in the units of the map; they are not forecasts of what will happen, or of how long it takes. The archetype thresholds are conventions that structure the comparison, and the robustness test shows how much the top pathways depend on the simulation settings.

> **Without a squashing function:** A stage that runs away is stopped and counts as not converged. If many pathways do not converge, an amber notice says so: the model is then unstable without squashing (see the Baseline Equilibrium tab), and a bounded transfer function is needed to compare pathways. The robustness test then compares with the hyperbolic tangent instead of varying the steepness.

14. Exporting Your Work

14.1 Export JSON / Import

Export JSON (in the header) downloads the entire model (every concept, relationship, scenario, and Advanced option) as a single .json file. Import loads one back and, after confirmation, replaces the current model. The file is checked first, and anything that had to be corrected is listed in a notice (Section 3.3). Although the model is also saved automatically in the browser (Section 3.3), Export JSON remains the way to keep a versioned record of a model as it evolves, share it with a collaborator, or move it to another computer. The model file is always the complete model, whatever the analysis scope.

To export the model as tables (concepts with definitions, relationships, adjacency matrix), use Export model tables on the Model editor tab (Section 4.2).

14.2 Per-chart exports and the figure panel

Most charts and tables (the network map, the Baseline Equilibrium convergence trace, the Scenarios comparison, the Sensitivity response table, the Transition Point curves and rankings, the Transition Pathways results, and the list of influence routes) have their own Export CSV or PNG button. While a partial scope is active, the downloaded file name includes the scope, for example "influence-routes-water-module.csv", so a result saved from a module view cannot later be mistaken for a full-nexus one.

The figure panel

The main charts of four tabs have a Customize & export button: the Baseline Equilibrium convergence plot; the Scenarios & Simulation comparison bar chart and trajectory charts; the Transition Point response and marginal-benefit curves; and the Transition Pathways effort-gain plane, outcome trajectory, and benefit accrual chart. It opens

the figure panel under the chart. While the panel is open, the chart on screen is resized to the exact export size, so what you see is what is saved.

Setting	What it does
Figure size	Five presets sized for A4 journal pages: Single-column portrait (84 by 110 mm), Single-column square (84 by 84 mm), Double-column landscape (174 by 100 mm), Full-page portrait (170 by 240 mm), and Full-page landscape (240 by 170 mm), or a custom width and height. The padlock keeps the aspect ratio.
Resolution and format	150, 300, or 600 dpi. PNG is a high-resolution raster with its physical size embedded, so a word processor places it at the right size; SVG is a vector file that can be edited; PDF is a single page.
Titles	The plot title and the axis titles. The title is written into the exported file, so the figure says what it shows.
Fonts	Separate sizes for the title, the axis titles, the tick labels, and the legend.
Legend and palette	Legend at the top, bottom, left, or right, or hidden. Palette: Default, Colorblind-safe (Okabe-Ito), or Grayscale.
Lines and points	Decimal places on the axes, gridlines on or off, and line thickness, marker size, and opacity.
Series shown and order	Tick which concepts or outcomes appear, and choose their order (as selected, alphabetical, or by a value where the chart has one). The effort-gain plane can also cap how many background points are drawn.
Reset appearance, Export figure	Reset appearance returns the fonts, gridlines, legend, palette, line settings, and decimals to their defaults. Export figure downloads the file.

The trajectory comparison on Scenarios & Simulation has three charts (baseline, scenario, and difference) that share one panel and are exported as three files. The network map and the Sensitivity Analysis charts keep their own PNG export, with the chart title written into the image. In Figure style the network map's Export PNG saves the whole map, cropped to its content, at three times screen resolution (Section 4.1).

14.3 Export Analysis Workbook

The Export Analysis Workbook button in the header builds one documented, multi-sheet .xlsx file covering the model and every analysis that has been run this session, with frozen header rows, autofilters, and auto-sized columns throughout. Its file name includes the scope when a partial scope is active. The transfer function of the simulation settings is recorded on the relevant sheets, including "none (linear, unbounded)". Monte Carlo results are not part of the workbook; use the CSV export on that tab.

How the scope is recorded

The first sheet, **Analysis Scope**, states which part of the system the workbook's analyses describe: the analysed network, the view mode, the modules in view, whether isolated concepts were included, how many concepts and relationships were analysed, and how to interpret subsystem or interface results. Three groups of sheets follow different rules:

- **The complete model:** Model Structure and Adjacency Matrix always contain every concept and relationship. When a partial scope is active, Model Structure has an "In Analysed View" column.

- **The analysed network:** every metric, baseline, scenario, sensitivity, transition point, route, loop, and chart sheet describes the network in the current scope.
- **Fixed scales:** Multi-scale Ranks and Leverage by Scale always cover the full nexus, each module on its own, and the nexus interface, whatever the scope.

Sheets always included

Sheet	Contents
Analysis Scope	The scope the workbook describes, and how to read it (see above).
Model Structure	Every concept: ID, name, module, definition, starting value, outgoing/incoming relationship counts (and In Analysed View under a partial scope).
Adjacency Matrix	The full concept × concept weight grid.
Network Metrics	Indegree, outdegree, centrality, net influence, driver score, receiver score, betweenness, role, influence score, and sensitivity score for every concept in scope (Section 6.2).
Category Metrics	The Section 6.6 aggregates per module.
Baseline Equilibrium	Per-concept starting value (as used by the run), equilibrium and change, plus a diagnostics block (initial state type, converged, iterations, residual, out of date). Shows a "not run yet" note if Baseline Equilibrium has not been run.
Scenarios Summary	Every scenario's name, type, description, derived intervention count, whether it has been run, convergence, iterations, and out-of-date status.
One sheet per scenario	Baseline / Scenario final values and effect size per concept, or a "not run yet" note.
Feedback Loops	Every feedback loop in scope with its modules, whether it lies within or between modules, length, type, and strength, headed by a summary for the whole system, within modules, and between modules.
System Overview Metrics	The Section 6.5 system-level statistics, including clustering, modularity, and interface relationships.
Multi-scale Ranks	Every concept's influence, centrality, and betweenness value and rank globally, in its own module, and in the nexus interface, with the "importance mainly from" reading.
Leverage by Scale	The five concepts with the highest influence score at every scale.
Influence Routes and Bridge Concepts	The routes for the selection currently set on the Influence Routes tab (strongest 5,000 listed), headed by a description of that selection, and the bridge-concept table. If the selection is not possible in the exported scope, a note explains why.

Sheets included when available

Sheet	Appears when…
Baseline Iterations	Baseline Equilibrium has been run: the full iteration-by-iteration trace behind the convergence plot.

Sheet	Appears when…
Sensitivity Analysis	Sensitivity Analysis has been run: elasticity table plus the tornado ranking.
Transition Point Analysis	Transition Point Analysis has been run: the full results table.
Pathway Analysis Setup	A pathway analysis has been run: the outcomes, levers, and options used, the simulation settings, the number of pathways per archetype, the invariance result, and whether the result is out of date.
Transition Pathways	A pathway analysis has been run: every pathway (the 5,000 with the highest total outcome gain, if there are more) with all its metrics, its archetypes or status, and whether it is on the efficient frontier.
Pathway Rankings	A pathway analysis has been run: the ten best pathways of each archetype.
Pathway Stages	A pathway analysis has been run: stage by stage, the top-ranked pathways and those on the efficient frontier, with the value of every outcome at the end of each stage.
Pathway Lever Profiles	A pathway analysis has been run: each lever on its own from Business-as-Usual, and why a lever was left out of the pathways.
Pathway Portfolios	A pathway analysis has been run: the opening moves, their regret, and the best continuation per archetype.
Pathway Invariance Test	A pathway analysis has been run: whether the order of the levers changed the end state.
Pathway Robustness	A pathway analysis has been run with the robustness test switched on.
Category Interaction Matrix	At least two modules exist: link-count and total-weight grids.
Convergence Diagnostics	Baseline Equilibrium has been run: the diagnostics in a compact metric/value layout.

Chart images

Five chart sheets are always created: Network Graph, Sensitivity Scores Chart, Baseline Convergence Plot, Scenario Comparison Plot, and Transition Point Curves. Each is embedded as an image if that chart has been generated at some point this session (charts you have visited stay cached for export even after navigating to another tab); otherwise the sheet holds a plain-language note explaining what to do to include it (usually: visit that tab, and run the relevant analysis). When a pathway analysis has been run, a sixth chart sheet, Pathway Effort-Gain Plane, holds the effort-gain chart of the Transition Pathways tab.

Out-of-date results in the exported workbook

If a result was computed before a later edit to the model or a change of scope, the workbook says so directly: a "Stale" row or column on the Baseline Equilibrium, Convergence Diagnostics, and Scenarios Summary sheets, a bold note at the top of any Sensitivity Analysis, Transition Point Analysis, or per-scenario sheet whose result predates the change, and a Stale row on the Pathway Analysis Setup sheet. This mirrors the amber banner shown in the app itself (Sections 8.5 and 9.7); re-run the affected analysis before exporting if you want the workbook to reflect the current model.

15. Glossary

Term	Definition
Concept	A variable or factor in the model, with an activation value from -1 to +1.
Relationship / Edge	A directed, signed, weighted causal link from one concept to another.
Weight	A relationship's strength and direction, from -1 (strongest suppressing) to +1 (strongest reinforcing).
Activation	A concept's current value on the -1..+1 scale; how suppressed or active it is.
A(0) / starting value	The initial activation vector: every concept's activation at the start of a run.
Squash function	The function (tanh, sigmoid, or trivalent) that maps incoming influence back onto the -1..+1 scale each iteration. With no squashing function, the values are left as they are (Section 2.2).
Converged	The run's largest per-concept change between iterations dropped below the convergence threshold.
Divergence (runaway)	Without a squashing function, a run in which activation grows without limit. It is stopped once any concept passes ±1000 and is reported as diverging.
Equilibrium	The state a run settles into once it converges (or the state at the iteration cap, if it does not).
Module	A group of concepts sharing the same category: a subsystem of the nexus.
Analysis scope	The part of the model every tab currently analyses: the full nexus, one or more modules, or the nexus interface.
Nexus interface	The relationships between concepts of different modules; the coupling between subsystems.
Hide / Focus	Network-tab display of concepts outside the analysis scope: removed (Hide) or faded (Focus).
Folded module	A module drawn as a single block on the Network tab. Folding changes only the map, never the model or the analyses.
Module overview / Module view	Network-tab displays with every module folded (overview), or with one module opened and the others folded around it (module view).
Bundled link	On the map, one arrow standing for every relationship between the same two blocks, labelled with their number and net weight.
Definition	A concept's description of what it means and how it is measured.
Driver	In scenarios: a concept forced to full activation (±1) and held fixed for a whole run.
Lock	Holding a concept's value fixed for a whole run, at any value (not only ±1).
Scenario	A named variant of the model with one or more interventions applied, compared against the baseline.

Term	Definition
Intervention	A driver/lock, an overridden starting value, or an adjusted relationship weight within a scenario.
Effect size	How far a concept's scenario equilibrium differs from its baseline equilibrium.
Synergy / Trade-off	Interventions combining to produce more effect than the sum of their parts (synergy), or improving one outcome while worsening another (trade-off).
Indegree / Outdegree	Sum of absolute weights of a concept's incoming / outgoing relationships.
Centrality	Indegree + outdegree.
Driver score / Receiver score	The share of a concept's connection weight that is outgoing / incoming.
Betweenness	The share of shortest directed routes between other concepts that pass through a concept.
Hub / Driver role / Receiver role / Connector / Isolated	The five structural roles a concept can be classified into; see Section 6.2 for the exact thresholds.
Influence score	Perturbation-based: the average movement a concept causes in every other concept when driven to +1 from an all-zero start.
Sensitivity score	Perturbation-based: the average movement a concept undergoes across every other concept's driven run.
Clustering coefficient	How often a concept's neighbours are also linked to each other, averaged over the network.
Modularity (Q)	How much more tightly the modules are connected internally than to each other, compared with chance.
Leverage point	Here: the concept with the highest influence score at a given scale.
Influence route	A chain of relationships along which a change in one concept can reach another.
Bridge concept / Bottleneck	A concept many routes pass through / one carrying at least half of the selected routes' combined strength.
Boundary crossing	A point where a route or loop moves from one module into another.
Feedback loop (reinforcing / balancing)	A route back to its own starting concept; reinforcing amplifies change (even number of negative links), balancing dampens it (odd number).
Elasticity	In Sensitivity Analysis, the average slope of an output across the full swept input range.
Monte Carlo simulation	Re-running the model many times with random noise added to the weights (and optionally the starting values), to see how widely the results spread.
Weight uncertainty (±)	The largest amount by which a Monte Carlo run may nudge a weight, up or down.
Random seed	The number that fixes a Monte Carlo run's random sequence, so the run can be repeated exactly.

Term	Definition
Percentile (P5, P95)	The value below which 5% (or 95%) of the runs fall; together they bound a plausible range.
Transition point	In Transition Point Analysis, the intervention intensity from which an outcome's marginal benefit declines most steeply: the point of diminishing returns. It is 1 when the marginal benefit does not decline before full intensity.
Marginal benefit	The change in an outcome (on the Transition Pathways tab, in the total outcome gain) per unit of intensity over one step of a sweep, counted in the direction that counts as benefit.
Share at TP	The change in an outcome up to its transition point as a share of its change over the whole sweep.
Efficiency	In Transition Point Analysis, total effect ÷ transition point.
Transition pathway	An ordered sequence of interventions (levers) that moves the system from Business-as-Usual towards a desired state, one stage at a time.
Stage	One step of a pathway: one lever implemented, after which the system settles before the next lever starts. An implementation phase, not a period of time.
Business-as-Usual (BAU)	On the Transition Pathways tab, the equilibrium of the model's own starting values with no scenario applied: the starting point of every pathway.
Nexus outcome	A concept a pathway should improve, with the direction that counts as better (higher or lower) and a weight.
Conditional transition point	A lever's transition point in the state a pathway has reached so far, rather than from Business-as-Usual.
TP shift	Conditional minus stand-alone transition point: negative when earlier stages make a lever's returns diminish sooner (enabling), positive when they delay them.
TOG / CII / PEI	Total Outcome Gain (improvement of the outcomes over BAU), Cumulative Intervention Intensity (total effort), and Pathway Efficiency Index (TOG ÷ CII).
EBI / IG / SS	Early Benefit Index (whether benefits arrive ahead of effort), Interaction Gain (gain beyond the levers' separate effects), and Synergy Score (number of outcomes improved).
Effort weight	How demanding one unit of a lever is compared with the others; enters the CII.
Pathway archetype	Low-Hanging Fruit, High-Impact, Efficiency, or Deep Investment: a type of pathway defined by rules on its metrics (Section 13.5).
Admissible pathway	A pathway that converged, has no trade-offs, and reaches the minimum gain; only these are sorted into archetypes.
Efficient frontier	The pathways for which no other pathway achieves more gain with less effort.
Opening / Regret	The first stage or stages of a pathway / how much total outcome gain an opening gives up compared with the best pathway of all.

Term	Definition
Path dependence	When the order of the levers, not only which levers, decides the state the system ends up in; checked by the invariance test.
Model check	The automatic list of likely mistakes and next steps shown on the Network and Model editor tabs.
Tab group	One of the four groups of tabs: Model building, State and scenario analysis, Sensitivity analyses, and Transition analyses.
Figure panel	The settings under Customize & export on a chart: size, resolution, fonts, palette, legend, and file format.
Out of date (stale)	A displayed or exported result that was computed before a later edit to the model or change of scope, and no longer reflects it.

16. Known Limitations and Troubleshooting

16.1 Prototype scope

- Runs entirely in the browser. The model is saved automatically in that browser only; analysis results are not saved. Use Export JSON to keep or move a model (Sections 3.3 and 14.1).
- Single-user: there is no multi-user collaboration or server-side storage.
- Monte Carlo (Section 11) perturbs weights and starting values only, independently and with one spread for all; it does not vary the structure of the map, and per-relationship uncertainty ranges are not offered. Its results are not saved and are not part of the Analysis Workbook.
- Without a squashing function most maps run away; the stopping value of ±1000 is a safeguard, not a result (Section 2.2).
- Interface details: numeric fields snap back to 0 when you clear them, so type over the value instead of deleting it; the Monte Carlo settings return to their defaults when you leave the tab; the header bar is not designed for phone-sized screens.
- No direct DOCX or PPTX export, and no PDF of a whole analysis, from within the tool: use Export Analysis Workbook (.xlsx), or the per-chart CSV exports and figure exports (PNG, SVG, PDF), and bring the results into another document.
- Undo covers concepts and relationships, not scenarios or settings.
- Transition Pathways: the number of pathways grows very fast with the number of levers and stages. Every order is limited to 20,000 pathways; beyond that, use beam search or fewer stages. With robustness switched on, the analysis takes about five times as long.
- Very large models (roughly 100+ concepts) work, but some parts slow down: the Model editor tab renders a full dropdown of every concept for both ends of every relationship, and the multi-scale comparison on Influence Metrics takes one to two seconds at around 120 concepts. The Network tab remains the most practical way to work with a large model.

16.2 Frequently asked questions

“Why did my Baseline Equilibrium run converge straight to zero?”

Every concept was starting at 0 with nothing locked or forced, or the steepness λ is too low for the weights to sustain activation. The first is mathematically guaranteed for every squash function this tool offers (Section 2.2); give at least the currently active concepts a real starting value (in the Inspector, on the Model editor, or in the custom initial state on the Baseline Equilibrium tab). For the second, try a higher λ on the Baseline Equilibrium tab.

“A concept disappeared from the map.”

It is probably outside the current analysis scope: its module is not selected, or it has no relationship inside the scope while "Show isolated concepts" is off. Click Back to full nexus in the scope bar, switch the map to Focus, or click its name in the Model check.

“Why are my results different for a module than for the full nexus?”

Because a module analysed on its own has the links to the rest of the system cut (Section 5.5). The difference is the effect of those links. The multi-scale comparison (Section 6.4) shows the same contrast for every concept at once.

“Why did my Influence/Sensitivity scores not change when I edited a concept's starting value?”

By design: Section 6.3 explains that these scores are always computed from an explicit, forced all-zero start (except the one concept being driven), independent of the starting values set on the model. They measure the network's structure alone, not its current state.

“The app or my exported workbook says a result is out of date. What does it mean?”

The result was computed before a later edit to the model, a scenario's interventions, or the analysis scope (Sections 8.5, 9.7, 14.3). The numbers are not wrong, only out of date. Re-run that analysis to refresh them.

“A chart sheet in my exported workbook shows a note instead of an image. Why?”

That chart was never drawn in the browser this session, usually because its tab was never opened or the analysis feeding it was never run (Section 14.3). Visit that tab (and run the analysis, if it needs one) before exporting again.

“I closed the tab and my model is gone.”

Normally it is restored automatically on your next visit (Section 3.3). It will not be there if you used a different browser or computer, a private window, or cleared the browser's site data. In that case, use Import with your most recent Export JSON file.

“I replaced my model by accident.”

Press Ctrl+Z (Undo) before reloading the page to bring back the previous concepts and relationships. Next time, choose Download a copy first, then replace in the dialog.

“Nothing happens when I choose Blank model or an example.”

In version 0.2 and earlier, some browsers blocked the confirmation question and the model was never replaced. From version 0.2.1 the question is a dialog inside the tool. If you still see no dialog, reload the page so the browser loads the current version.

“Why are some concepts missing from the map?”

Their module may be folded (Section 4.1): choose Full model in the structure switch, or double-click the block. Otherwise they are outside the analysis scope (see the question about a concept that disappeared from the map).

“On Transition Pathways, an archetype says No pathway meets all conditions.”

The archetypes are defined relative to the other pathways of the same run, and each needs several conditions at once. The card lists how many pathways meet each condition on its own, so you can see which one nobody meets. In a model where a single lever already moves the whole system, for example, the most efficient pathways all need

very little effort, fall into Low-Hanging Fruit, and leave Efficiency empty. That describes the model, not a failure of the analysis (Section 13.5).

“Many pathways did not converge.”

Every stage is placed at a transition point, where the model is still responding strongly and settles slowly, and some lever combinations make the model oscillate instead of settling. Such pathways are reported but not sorted into archetypes. Raising Maximum iterations per run helps when runs settle slowly; if they oscillate, it does not, and the balancing feedback loops through the oscillating concepts are worth examining (Section 7.5).

“Why is a lever’s transition point on Transition Pathways different from the one on Transition Point Analysis?”

Both use the same definition, the intensity from which the marginal benefit declines most steeply, but on different curves. Transition Pathways uses the total gain in the nexus outcomes, in the direction you marked as better, swept from the state the pathway has reached (from BAU for the lever profiles). Transition Point Analysis looks at one outcome at a time, in the direction that outcome moves (Sections 12 and 13.3).

“I switched off the squashing function and the model diverges.”

Without squashing nothing bounds activation, so any feedback that reinforces itself makes the values grow on every pass. The tool stops the run when a concept passes ±1000 and reports Divergence (Section 2.2). This is the model's real linear behavior, which the squashing function normally hides. Weaken or rebalance the reinforcing loops (Section 7.5), or switch a bounded function back on.

“Why do so many Monte Carlo runs not settle when the baseline does?”

The nominal run only shows that one set of weights settles. A model that settles slowly (the water-energy-food example needs 73 iterations) or lies near an oscillation can be pushed beyond the iteration limit, or into a cycle, by small changes in the weights. Raise Max iterations or lower the weight uncertainty, and treat the share that does not settle as a finding about the model (Section 11.3).

“Can I repeat a Monte Carlo run exactly?”

Yes: use the same model, settings, and seed (Section 11.4).

“How do I get an arrow out of a tangle on the Network tab?”

Select the relationship and drag its label sideways to bow the line, or drag one of the two handles at its ends to connect it to another concept or dot (Section 4.1). Hover a concept to fade everything except its own connections, and use Labels to show the arrows by color and width only.

“A tab says Something went wrong.”

Click Try again, or switch to another tab and back. Your model is not affected. If it keeps happening, use Export JSON to save the model and report the problem.

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Prototype scope: runs entirely in your browser; the model is saved automatically in this browser only. PDF/DOCX/PPTX export of a whole analysis is not yet implemented.